

CHRONOSCOPE USER MANUAL

This document details the methodological foundations, usage, and parameter configuration for the core analysis modules integrated into the ChronoScope GUI.

TABLE OF CONTENTS:

0. MODULE 0: PREPROCESSING (preprocessing.py)	1
0.1. Outlier Removal	1
0.2. Detrending	1
0.3. Smoothing	2
1. MODULE I: RHYTHM ANALYSIS (rhythm_analysis.py)	2
1.1. Python-JTK (JTK_CYCLE)	3
1.2. AR-JTK (AutoRegressive JTK)	4
1.3. Cosine - Kendall	6
1.4. Cosinor (OLS)	7
1.5. Harmonic Cosinor	8
1.6. FOURIER F24 (Wijnen et al., 2006)	9
1.7. LOMB-SCARGLE Periodogram	10
1.8. Spectral Analysis (Periodogram)	10
1.9. Continuous Wavelet Transform (CWT)	12
1.10. Linear Mixed Effects (LME)	13
2. MODULE II: CosinorPy (cosinor_analysis.py)	15
2.1. Periodogram Analysis	15
2.2. Cosinor (independent data)	16
2.3. Cosinor (dependent/population data)	19
2.4. Compare conditions (independent data)	20
2.5. Compare conditions (dependent/population data)	23
2.6. Nonlinear cosinor (independent data)	25
2.7. Nonlinear cosinor (dependent/population data)	28
2.8. Nonlinear compare (independent data)	30
2.9. Nonlinear compare (dependent/population data)	33
3. MODULE III: CircaCompare (circacompare_analysis.py)	37
3.1. Single Fit (Robust Fitting)	37
3.2. Compare Groups (Differential Analysis)	38
4. MODULE IV: RhythmCount (rhythmcoun analysis.py)	39
4.1 Fit Single Model	39
4.2 Fit All Models	40
4.3 Fit Best Model (Automatic Model Selection)	41
4.4 Parameter Confidence Intervals	42
4.5 Group Comparison	43
5. MODULE V: AI CONSENSUS – CONSENSUS RHYTHMICITY SCORE	44
6. MODULE VI: Locomotor Activity Analysis (DAM/AWD data)	45
6.1. Activity Profile	46

0. MODULE 0: PREPROCESSING (preprocessing.py)

Raw circadian time series often contain noise, measurement artifacts, and slow baseline drifts that are unrelated to the underlying rhythm. Applying preprocessing before analysis can improve the accuracy of estimated parameters and reduce false-positive or false-negative rhythm detection.

0.1. Outlier Removal

- Purpose: Identifies and removes data points that deviate anomalously from the rest of the time series. Outliers are replaced with `NaN` rather than deleted, preserving the time axis for downstream stages.
- Available methods:
 - IQR (Interquartile Range):
Mechanism: Computes the first (Q1) and third (Q3) quartiles of the series. Any value outside the interval $[Q1 - k \times IQR, Q3 + k \times IQR]$, where $IQR = Q3 - Q1$, is flagged as an outlier.
Key parameter Threshold (k): Multiplier applied to the IQR. Lower values are more aggressive.
 - Z-score:
Mechanism: Computes the mean (μ) and standard deviation (σ) of the series. Any value with $|z| > \text{threshold}$, where $z = (x - \mu) / \sigma$, is flagged.
Key parameter Threshold: Number of standard deviations.

0.2. Detrending

- Purpose: Removes slow, non-rhythmic trends from the signal. Uncorrected trends inflate the estimated amplitude and bias the acrophase, because the statistical model attributes part of the trend energy to the fitted sinusoid.
- Available methods:
 - Moving Average Subtraction
Mechanism: Computes a centered rolling mean with a window equal to one expected period (default 24 h), then subtracts it from the raw signal. Because the window spans exactly one cycle, the rolling mean tracks the slow baseline without capturing the oscillation itself.
Key parameter Window: Width of the centered rolling mean in hours.
 - Linear
Mechanism: Fits a least-squares regression line to the entire time series (using only valid, non-`NaN` timepoints) and subtracts it.
 - Polynomial
Mechanism: Fits a polynomial of degree n to the time series using least-squares and subtracts the polynomial trend. Degree 2 (quadratic) handles U-shaped or inverted-U-shaped baselines; degree 3 adds one additional inflection point.
Key parameter Degree: Polynomial degree (1–5).

0.3. Smoothing

- Purpose: Attenuates high-frequency noise while preserving the shape and timing of the circadian oscillation.
Unlike detrending, smoothing is a low-pass operation: it keeps low frequencies (slow oscillations) and attenuates high frequencies (noise). Detrending is a high-pass operation: it removes low frequencies (trend) and keeps higher frequencies (rhythm + noise).
All smoothing methods interpolate `NaN` values (introduced by outlier removal) before filtering and restore them as `NaN` afterwards, so the time axis structure is preserved.
- Available methods:
 - Moving Average:
Mechanism: Replaces each sample with the average of a centered window of w samples.
Key parameter Window: Number of samples in the centered rolling window.
 - Savitzky-Golay:
Mechanism: Fits a local polynomial (degree 3) to each window and uses the fitted value at the window center. This is mathematically equivalent to a weighted moving average where weights are derived from the polynomial fit, giving more emphasis to the center of the window and preserving peak curvature better than a simple mean.
Key parameter Window: Number of samples. Must be odd and ≥ 5 (enforced automatically).
 - Butterworth:
Mechanism: Designs a Butterworth infinite impulse response (IIR) low-pass filter and applies it twice in opposite directions (`filtfilt`). The two-pass application cancels phase distortion, so the acrophase of the filtered signal is identical to the original — this is called zero-phase filtering. The filter passes frequencies below the cutoff and attenuates frequencies above it with a maximally flat response in the passband (no ripple).
Key parameter Order: Filter order. Controls rolloff steepness.
Key parameter Cutoff: Low-pass cutoff expressed as a period in hours. Oscillations shorter than this period are attenuated.

1. MODULE I: RHYTHM ANALYSIS (rhythm_analysis.py)

This module provides a broad suite of algorithms for rhythm detection, ranging from classical parametric methods to robust non-parametric tests. It is designed for initial screening of rhythmicity in diverse datasets.

Note for 1.1, 1.2 and 1.3 - Empirical p-values (Hutchison et al., 2015): For JTK-based methods (Python-JTK, AR-JTK, and Cosine-Kendall), ChronoScope

supports empirical p-value estimation via permutation testing. This approach constructs a null distribution of the maximum $|\tau|$ statistic across all waveform templates under random data, and estimates the p-value as the fraction of permutations where this null statistic equals or exceeds the observed value. Unlike analytical p-values, empirical p-values are correctly calibrated for small sample sizes and do not require additional correction for the multiple template comparisons inherent to JTK-style methods. Reference: Hutchison AL, Maienschein-Cline M, Chiang AH, et al. (2015) Improved statistical methods enable greater sensitivity in rhythm detection for genome-wide data. PLoS Computational Biology, 11(3): e1004094.

1.1. Python-JTK (JTK_CYCLE)

- Purpose: The gold standard for detecting rhythmicity in datasets with non-normal distributions, outliers, or low sampling resolution.
- Mechanism: A non-parametric test that compares the rank order of data against a series of reference triangle templates aligned to real timepoints.
- Key Parameters:
 - Period Range: The window of cycle lengths to test (e.g., 20-28h).
 - Asymmetry: Defines the skewness of the reference template (default 0.5 for symmetric). Impact: Changing asymmetry helps detect "sawtooth" patterns (fast rise, slow decay).
 - Asymmetry Search Range: When enabled, automatically tests waveform asymmetries from 0.2 to 0.8 in steps of 0.1 (7 values) to find the best-fitting waveform shape, following the empirical JTK_CYCLE approach of Hutchison et al. (2015). This is equivalent to the widths_02-22.txt parameter file in the original eJTK implementation. When disabled (default), only the single fixed asymmetry value above is used.
 - P-value method: Controls how the significance p-value is estimated.
 - Analytical (faster): Uses the asymptotic Kendall tau distribution (standard). May be anti-conservative (optimistic) when sample sizes are small ($n < \sim 12$ timepoints per cycle).
 - Empirical (permutation): Builds a null distribution by permuting the data, following Hutchison et al. (2015). The null distribution is the maximum $|\tau|$ across all period/lag/asymmetry templates under random data — this correctly accounts for the multiple template comparisons within JTK without further correction. More accurate for small samples but substantially slower.
 - Permutations: (visible only when Empirical is selected) Number of random permutations used to build the null distribution. Default: 1000. Minimum recommended: 500 for exploratory analyses; ≥ 1000 for publication-quality results.
- Output Interpretation (Results Table):

- amplitude: The estimated amplitude of the rhythm, calculated non-parametrically as half the distance between the 90th and 10th percentiles of the data $(p_{90} - p_{10}) / 2$.
- acrophase: The estimated time of the peak in hours. It is derived from the best-fitting lag, adjusting for the asymmetry of the template and the sign of the tau correlation.
- tau: The Kendall tau correlation coefficient between the ranked data and the best-fitting ranked template. It indicates the strength of the fit (values range from -1 to 1). A negative tau indicates the data is inversely correlated with the template (i.e., the rhythm peaks where the template is at its minimum); the acrophase is automatically corrected for this.
- lag: The time offset (in hours) of the best-fitting template relative to time zero. It represents the raw phase shift before the acrophase correction is applied.
- p (raw): The raw p-value from the Kendall tau correlation test. It represents the statistical significance of the best-fitting template without any correction for multiple testing.
- p (BH adj): The Benjamini-Hochberg adjusted p-value. This controls the False Discovery Rate (FDR) and is less conservative than Bonferroni. It is the recommended metric for determining significance, especially when analyzing multiple variables simultaneously, as it provides better statistical power than Bonferroni while controlling the FDR.
- p (Bonf adj): The Bonferroni-adjusted p-value. This is a very conservative correction that controls the Family-Wise Error Rate (FWER). If this is < 0.05 , the rhythm is highly significant, but it may miss weaker true rhythms (Type II error).
- $p_value / p (BH adj) / p (Bonf adj)$ in Empirical mode: When "Empirical (permutation)" is selected, all three p-values report the same empirical estimate. The permutation-based null distribution already corrects for multiple template comparisons internally, so no additional Bonferroni or BH correction is applied at the within-series level. For genome-wide analyses, apply BH correction across genes using the raw empirical p-values.

1.2. AR-JTK (AutoRegressive JTK)

- Purpose: Designed for time-series data where sequential points are correlated (red noise).
- Mechanism: Automatically tests rank residuals for white noise using the Ljung-Box test. If red noise is detected, it pre-whitens the residuals using an AutoRegressive (AR) model before applying the standard JTK algorithm.
- Key Parameters:

- AR Lag: The number of previous time points influencing the current point. Impact: Higher lags correct for longer-term dependencies but require longer time series.
- Force pre-whitening: When unchecked (default), AR pre-whitening is applied only if the Ljung-Box test detects autocorrelation in the rank residuals. When checked, the Ljung-Box test is skipped and AR pre-whitening is always applied regardless of the detected noise structure. Use this option when prior knowledge of the data confirms temporal autocorrelation, or to ensure consistent behavior across datasets.
- Asymmetry Search Range: When enabled, automatically tests waveform asymmetries from 0.2 to 0.8 in steps of 0.1 (7 values) to find the best-fitting waveform shape, following the empirical JTK_CYCLE approach of Hutchison et al. (2015). This is equivalent to the widths_02-22.txt parameter file in the original eJTK implementation. When disabled (default), only the single fixed asymmetry value above is used.
- P-value method: Controls how the significance p-value is estimated.
 - Analytical (faster): Uses the asymptotic Kendall tau distribution (standard). May be anti-conservative (optimistic) when sample sizes are small ($n < \sim 12$ timepoints per cycle).
 - Empirical (permutation): Builds a null distribution by permuting the data, following Hutchison et al. (2015). The null distribution is the maximum $|\tau|$ across all period/lag/asymmetry templates under random data — this correctly accounts for the multiple template comparisons within JTK without further correction. More accurate for small samples but substantially slower.
Note: When Empirical is selected, permutations are performed on the pre-whitened series (when AR correction is applied), ensuring that the null distribution reflects the actual noise structure of the data after autocorrelation removal.
- Permutations: (visible only when Empirical is selected) Number of random permutations used to build the null distribution. Default: 1000. Minimum recommended: 500 for exploratory analyses; ≥ 1000 for publication-quality results.
- Output Interpretation (Results Table):

Note: The output table shares the same metrics as standard Python-JTK (p_value, bonf_p_value, bh_p_value, period, acrophase, tau, lag, asymmetry, n_tests). However, these metrics come from two different internal analyses:

 - Statistical metrics (p_value, bonf_p_value, bh_p_value, n_tests): Derived from the JTK run on the pre-whitened series. They represent rhythm significance after removing autocorrelated noise, and are

generally more reliable than standard JTK metrics when dealing with high-resolution or continuous sampling data.

- Phase and shape metrics (acrophase, tau, lag, asymmetry, period): Derived from the initial JTK run on the original raw data. This ensures that phase estimation is not distorted by the pre-whitening step.
- amplitude: Recalculated from the original raw data using the 90th–10th percentile range, ensuring the biological magnitude remains interpretable regardless of whether pre-whitening was applied.
- AR correction status: The result includes a status message indicating whether AR pre-whitening was actually triggered, either because the Ljung-Box test detected autocorrelation, or because Force pre-whitening was enabled. If no autocorrelation was detected and Force pre-whitening is off, the result is equivalent to standard Python-JTK.
- n_tests: The number of period × lag × asymmetry combinations tested in the pre-whitened JTK. Used for the multiple testing corrections.
- Empirical p-values and AR pre-whitening: When both "Force pre-whitening" and "Empirical (permutation)" are active, the permutation null distribution is built from the pre-whitened series, not the original data. This ensures statistical validity: the null distribution and the observed statistic are computed under the same noise model.

1.3. Cosine - Kendall

- Purpose: A non-parametric alternative to standard Cosinor that evaluates rhythmicity using rank-order correlation rather than least-squares fitting.
- Mechanism: Computes the Kendall tau correlation between the ranked data and a ranked standard cosine wave template.
- Key Parameters:
 - Period Range: The window of cycle lengths to test (e.g., 20-28h).
 - Resolution: The step size of the phase-lag sweep in hours. Smaller values produce a finer-grained search across candidate acrophases, yielding a more precise phase estimate at the cost of computation time. Default: 0.5h.
 - P-value method: Same as Python-JTK. "Analytical (faster)" uses the Kendall tau asymptotic distribution; "Empirical (permutation)" builds a permutation null distribution of the maximum $|\tau|$ across all period/lag combinations (Hutchison et al., 2015). Note: Cosine-Kendall templates are symmetric by definition, so no asymmetry search option is available for this method.
 - Permutations: (visible only when Empirical is selected) Same as Python-JTK. Default: 1000.
- Output Interpretation (Results Table):

- `p_value` / `bonf_p_value` / `bh_p_value`: These represent the statistical significance of the rhythm based on the Kendall tau correlation, exactly as in the standard JTK method. `bh_p_value` is generally the recommended metric for significance.
- `amplitude`: Since this is a non-parametric method, the amplitude is estimated robustly using the data's percentiles: $(p_{90} - p_{10}) / 2$.
- `acrophase`: The time of the peak in hours. Note: The algorithm automatically corrects this value if the best correlation (`tau`) is negative. A negative `tau` indicates that the data peaks where the template is at its minimum, so the true acrophase is shifted by half a period ($\text{period} / 2$).
- `tau`: The Kendall tau correlation coefficient between the ranked data and the ranked cosine wave.
- `asymmetry`: This will always display as 0.5 in the results table, as standard cosine waves are perfectly symmetric by definition.
- `lag`: The raw phase shift of the best-fitting template before any negative-tau correction is applied.

1.4. Cosinor (OLS)

- **Purpose:** The traditional parametric method for assessing circadian parameters (MESOR, amplitude, and acrophase) in normally distributed data.
- **Mechanism:** Fits a regression model (Ordinary Least Squares) using cosine and sine transformations of time.
- **Key Parameters:**
 - **Search Mode:**
 - **Fixed Period:** Analyzes at a single user-specified period (no multiple testing correction needed; `adj_p_value` = `p_value`).
 - **Optimize Period:** Tests all periods in the range and selects the best fit by lowest AIC. Only the winning period is shown in the results table (not all tested periods).
- **Output Interpretation (Results Table):**
 - `p (raw)`: The raw F-test p-value from the OLS regression. Tests the null hypothesis that both the cosine and sine coefficients are simultaneously zero (i.e., no rhythm). This is the unadjusted significance of the fit at the selected period.
 - `p (Bonf adj)`: The Bonferroni-corrected p-value ($\text{raw } p \times \text{number of periods tested}$). In Optimize Period mode, this corrects for the multiple model comparisons performed across the period range. In Fixed Period mode, only one period is tested, so this equals the raw p-value.
 - `p_amplitude`: The t-test p-value for the cosine coefficient (β_{cos}), testing whether the cosine component is individually significant.
 - `p_acrophase`: The t-test p-value for the sine coefficient (β_{sin}), testing whether the sine component is individually significant.

- mesor: Midline Estimating Statistic Of Rhythm. This is the rhythm-adjusted mean (the intercept of the regression model), which is more accurate than a simple arithmetic mean if the sampling is uneven.
- amplitude: The rhythm amplitude (half the peak-to-trough distance), calculated mathematically from the cosine and sine beta coefficients: $((\beta_{\cos})^2 + (\beta_{\sin})^2)^{1/2}$.
- amplitude_ci: The 95% Confidence Interval for the amplitude. It is calculated using the delta method from the covariance matrix of the model parameters. If the lower bound is ≤ 0 , the rhythm's amplitude is not statistically significant.
- acrophase_hours / acrophase_rad: The time of the peak, provided in both hours and radians. It is derived using the arctangent of the beta coefficients.
- acrophase_ci: The 95% Confidence Interval for the phase shift in radians.
- period: The specific period from the tested range that yielded the best fit (lowest AIC).

1.5. Harmonic Cosinor

- Purpose: Detecting rhythms that deviate from a perfect sine wave (e.g., square waves, bimodal peaks).
- Mechanism: Fits a regression model composed of the fundamental frequency plus higher-order harmonics (multiples of the fundamental frequency).
- Key Parameters:
 - Period Range (Min / Max / Step): The range of cycle lengths to evaluate. The algorithm tests all periods in this range and selects the one that produces the highest R^2 .
 - n_harmonics: Number of cosine waves to sum. Impact: $n=1$ is standard cosinor; $n=2$ detects bimodal rhythms. High n values may overfit noise.
- Output Interpretation (Results Table):
 - p (F-test): The overall significance of the harmonic model from an F-test (H_0 : all coefficients except the intercept are zero). This is the raw p-value with no multiple testing correction.
 - period: The fundamental cycle length (in hours) that provided the best fit (highest R^2).
 - n_harmonics: The number of harmonics requested and evaluated in the model.
 - Primary Amplitude (H1): Amplitude of the dominant harmonic component (largest).
 - Primary Acrophase H1 (h): Peak time (hours) of the dominant harmonic.

- All Amplitudes: List of amplitudes for all harmonics, sorted descending by magnitude (H1 is the strongest).
- All Acrophases (h): Corresponding peak times for each harmonic, in the same order as All Amplitudes.
- Notes: If $n_harmonics > 1$, a secondary F-test compares the full model against a simple 1-harmonic cosinor. If the extra harmonics do not significantly improve the fit ($p > 0.05$), a warning appears here indicating the data may be unimodal. Use this as a diagnostic to avoid overfitting.

1.6. FOURIER F24 (Wijnen et al., 2006)

- Purpose: An effect-size measure rather than a significance test. Scores the strength of the 24h rhythm relative to background noise.
- Mechanism: Computes the power at the target frequency from the Fourier spectrum of the data, then divides it by the mean power obtained from a set of random permutations of the same data. The permutation baseline estimates the noise floor, making F24 a signal-to-noise ratio rather than a p-value.
- Key Parameters:
 - Target Period: The specific period to evaluate (usually 24.0h).
 - $n_permutations$: Number of random shuffles used to estimate the noise baseline.
- Output Interpretation (Results Table):
 - $f24_score$: The F24 rhythmicity index — the ratio of power at the target period to mean random power. Because this is an effect-size measure, researchers typically apply empirical thresholds (e.g., $F24 > 2.0$ or $F24 > 3.0$) to classify a signal as rhythmic.
 - $dominant_period$: The period (in hours) with the highest absolute power in the full spectrum. May or may not coincide with the target period.
 - $dominant_power$: The maximum absolute power value in the spectrum, corresponding to the dominant period.
 - $target_period$: The specific cycle length that was tested (e.g., 24.0 h).
 - $target_power$: The absolute spectral power measured at the target period.
 - $correlation_r$: The correlation between two replicates' complex Fourier coefficients at the target frequency, measuring phase and amplitude consistency across biological replicates. Populated automatically when the dataset contains a replicate or subject column with at least two unique values; empty otherwise.

1.7. LOMB-SCARGLE Periodogram

- Purpose: Detects dominant periodic signals in time series data, including datasets with irregular or uneven sampling intervals where standard FFT-based methods are not applicable.
- Mechanism: Computes a normalized Lomb-Scargle periodogram (Lomb, 1976; Scargle, 1982) using `scipy.signal.lombscargle`. The data is mean-centered before analysis. Power is evaluated across a grid of candidate periods and normalized to the range [0, 1] relative to data variance. The False Alarm Probability (FAP) for the dominant peak is estimated using the approximation $FAP \approx 1 - (1 - e^{-\text{power}})^M$, where M is the number of data points (a simplified proxy for the number of independent frequencies).
- Key Parameters:
 - Period Range: The (min, max) interval of periods to evaluate, in hours.
 - Oversampling Factor (internal): Number of evenly spaced candidate periods to test within the range (default: 1000).
 - Significance Level (α): The FAP threshold used to classify the dominant peak as significant or not (options: 0.05, 0.01, 0.001). The result in the Notes column will read "significant" if $FAP < \alpha$, and "not significant" otherwise.
- Output Interpretation (Results Table):
 - `dominant_period`: The period (in hours) that achieved the highest normalized power across the evaluated range.
 - `dominant_power`: The normalized power at the dominant period (range 0–1). Higher values indicate a stronger periodic signal relative to data variance.
 - FAP: False Alarm Probability for the dominant peak — the probability that the observed power level could arise by chance in a non-rhythmic signal. Lower values indicate higher confidence ($FAP < 0.05$ is a common threshold). Note that the FAP estimate uses a simplified approximation and should be interpreted accordingly.

Notes: Summary of the result: FAP value, significance verdict relative to the chosen α , and dominant period.

1.8. Spectral Analysis (Periodogram)

- Purpose: Identifies the dominant periodic signal and, optionally, all statistically significant periods across the spectrum. Useful for broad exploratory analysis when the rhythm period is unknown.
- Mechanism: Computes a power spectrum using one of three methods selected by the user. Significance is assessed using the Refinetti et al. (2007) threshold: $T = (1 - (\alpha/N)^{1/(N-1)}) \times \Sigma(\text{power})$, where N is the number of

observations and $\alpha = 0.05$. The threshold is computed from the full unfiltered spectrum before any Max Period cutoff is applied.

- Key Parameters:
 - Periodogram Type: Selects the spectral estimation method:
 - per — Standard FFT periodogram. Requires evenly sampled data; duplicate timepoints are collapsed by median.
 - welch — Welch's averaged periodogram. Reduces noise by averaging overlapping segments; also requires even sampling.
 - lombscargle — Lomb-Scargle variant. Handles uneven sampling without interpolation.
 - Max Period: Upper boundary (in hours) for the period axis. Frequencies corresponding to periods beyond this value are excluded from the output.
 - Detrend: When checked (default), the signal mean is subtracted before spectral estimation (DC offset removal). This prevents the zero-frequency component from inflating the spectrum and masking rhythmic peaks. Uncheck only if the absolute baseline level is meaningful for the analysis.
 - Find Prominent Peaks: When unchecked (default), only the dominant period (the frequency bin with maximum power) is reported. When checked, all spectral peaks whose power strictly exceeds the Refinetti threshold are identified, ranked by peak height in descending order, and reported as significant periods.
- Output Interpretation (Results Table):
 - dominant_period: The cycle length (in hours) with the highest spectral power across the evaluated spectrum.
 - Max Power: The raw spectral power value at the dominant period. Units depend on the selected method (power spectral density for per/welch; unnormalized power for lombscargle). Useful for comparing relative peak strength within the same analysis.
 - Threshold (Refinetti): The minimum power value required for statistical significance, computed dynamically from the data using the Refinetti et al. (2007) formula at $\alpha = 0.05$.
 - Significant Peaks (h): The periods (in hours) of all spectral peaks exceeding the threshold, listed in descending order of peak height. Only populated when Find Prominent Peaks is checked; otherwise shown as N/A.
 - Notes: Summary of the result — either the dominant period (default mode) or the count of significant peaks found (prominent peaks mode).

1.9. Continuous Wavelet Transform (CWT)

- Purpose: Analyzing non-stationary signals where the period or amplitude changes over time (e.g., damping rhythms, frequency modulation).
- Mechanism: Detrends the signal (mean subtraction), then convolves it with a wavelet function across a range of scales to generate a scalogram of power over time and period. Requires uniformly sampled data; if the input is unevenly sampled, it is linearly interpolated to a uniform grid before analysis. The internal period resolution is fixed at 0.25 h regardless of the Step parameter (which does not apply to CWT).

When the dataset contains a replicate or subject column with at least 2 unique values, CWT is run independently for each subject. The global wavelet spectra (power averaged over time at each scale) are then averaged across subjects, and `dominant_period` and `mean_power` are derived from this group-averaged spectrum. `period_variation` and `amplitude_modulations` are computed as the mean of the per-subject values. The scalogram visualization shows the first subject as a representative. When no subject column is present, the analysis runs on the cross-sectional mean at each timepoint.

- Key Parameters:
 - Period Range: Min/Max period (in hours) used to construct the frequency scales. Note: if the time series is too short to complete at least 3 cycles of the maximum period, the upper bound is automatically reduced to $\text{total_duration} / 3$. The lower bound is constrained to $2 \times \text{sampling_interval}$.
 - Wavelet Type: The shape of the probing function. Morlet (cmor1.5-1.0) — complex Morlet with bandwidth parameter 1.5 and center frequency 1.0 — is the standard choice for biological rhythms. Ricker (Mexican Hat) is an alternative real-valued wavelet.
 - Sampling Interval: Time between data points in hours. When set to Auto-detect (default), the interval is estimated from the median of consecutive time differences. When a value is specified manually, that value takes priority (except when the data was non-uniform and had to be resampled, in which case the new grid interval is used).
- Output Interpretation (Results Table):
 - `dominant_period`: The period (in hours) with the highest average power across the entire recording, derived from the global wavelet spectrum (power averaged over time for each scale).
 - `mean_power`: The mean of the global wavelet spectrum across all evaluated periods. Reflects the overall power density of the signal — not specific to the dominant period, but useful for comparing rhythmic strength across conditions.
 - `period_variation`: Standard deviation (in hours) of the instantaneous dominant period over time. At each timepoint, the period with maximum power is identified; this field measures how much that period drifts.

Values near 0 indicate a stable clock; larger values indicate period drift or frequency modulation.

- amplitude_modulations: Number of distinct local maxima detected in the time-varying amplitude envelope. The envelope is computed by averaging power across all scales at each timepoint. This quantifies recurrent amplitude fluctuations such as those seen in damping rhythms or beating patterns.
- Notes: Summary including dominant period, period variation, and amplitude modulations. For multi-subject analyses, also reports the number of subjects processed (e.g., Multi-subject CWT (n=4)).

1.10. Linear Mixed Effects (LME)

- Purpose: Ideal for studies with repeated measures across subjects or biological replicates, controlling for intra-group variance. Supports optional fixed-effect covariates (e.g., genotype, treatment batch) that are adjusted for while testing rhythmicity. Requires dependent data: each grouping unit (subject, replicate) must be measured at multiple time points so that the random intercept is identifiable.
- Mechanism: Fits a cosinor linear mixed-effects model with fixed rhythmic components ($\cos(2\pi t/T) + \sin(2\pi t/T)$), an optional set of user-specified covariates, and a random intercept for the grouping variable:

$$y = [\text{MESOR} + \text{covariates}] + \beta_{\cos} \cdot \cos(2\pi t/T) + \beta_{\sin} \cdot \sin(2\pi t/T) + u_{\text{group}} + \varepsilon$$
Rhythm significance is assessed via a Likelihood Ratio Test comparing the full cosinor model against a null model that retains the same covariates and random intercept but omits the rhythmic terms. Because both models include the same covariates, the LRT always has 2 degrees of freedom regardless of the number of covariates specified.
- Key Parameters:
 - Period: Target period in hours. If a range is specified (min/max/step), the model is fit independently at each period value and results are shown as separate rows — analogous to how JTK handles period scanning. The period with the lowest LRT p-value is identified as "Best Period."
 - Fixed Effects: Optional list of categorical or continuous covariates to include as additional fixed-effect terms (e.g., genotype, batch). Each selected column is added to both the full and null models, so the LRT tests only rhythm presence while adjusting for the listed covariates. Including a covariate that explains a substantial fraction of between-subject variance will reduce Var (random) and improve AIC without changing the amplitude or acrophase estimates. Leave empty to fit the baseline cosinor mixed model.

- Random Effect: The column defining the grouping variable (e.g., replicate or subject_id). If the specified column is not found, the engine automatically searches for a suitable grouping column (replicate, subject, animal, id, sample, day).
- Output Interpretation (Results Table):
 - p (LRT): Overall rhythm significance, calculated via a Likelihood Ratio test comparing the full cosinor mixed model against a null model (no rhythmicity). Evaluated using a chi-squared distribution with 2 degrees of freedom (cos and sin terms). This holds regardless of the number of fixed-effect covariates specified, because both models include the same covariate terms.
 - MESOR: Rhythm-adjusted mean, representing the fixed-effects intercept. If the model encounters convergence issues (e.g., singular random-effects covariance resulting in an intercept near zero), the algorithm automatically substitutes the empirical data mean as a fallback.
 - Amplitude: Population-level rhythm amplitude derived from the fixed-effect coefficients: $((\beta_{\cos})^2 + (\beta_{\sin})^2)^{1/2}$.
 - Acrophase (h): Population-level peak time in hours, derived from the fixed-effect coefficients.
 - R² (marginal): Proportion of total variance explained exclusively by the fixed rhythmic components, excluding the variance attributable to individual group differences.
 - Var (random): Variance attributed to the grouping variable — i.e., baseline differences between subjects or replicates. When a covariate that explains between-subject differences is added to Fixed Effects (e.g., genotype), this value decreases substantially; compare models with and without the covariate using AIC to confirm the improvement.
 - Var (residual): Remaining unexplained variance (noise) after accounting for both the rhythm and group effects.
 - AIC: Akaike Information Criterion. Computed as $-2\ell + 2k$, where ℓ is the log-likelihood and k is the number of model parameters (fixed effects + random intercept variance + residual variance). Lower values indicate better fit; useful for comparing models across tested periods.
 - BIC: Bayesian Information Criterion. Computed as $-2\ell + k \cdot \ln(N)$, where N is the number of observations. Penalizes model complexity more strongly than AIC for large datasets. Lower values indicate better fit.
 - Best Period: Only shown when multiple periods are tested. Marks "Yes" for the period with the lowest LRT p-value.

2. MODULE II: CosinorPy (cosinor_analysis.py)

ChronoScope integrates the CosinorPy library (Moskon, 2021), which provides four internal sub-modules: `cosinor` (multi-component analysis), `cosinor1` (single-component with detailed statistics), `cosinor_nonlin` (non-linear generalized cosinor), and `file_parser` (data I/O). The eight analysis methods exposed in the GUI are described below.

Note on data types: CosinorPy distinguishes two data structures: independent data, where each time point represents a different biological unit (cross-sectional design, no subject column), and dependent/population data, where the same subjects are measured repeatedly (longitudinal design, requires a subject column). ChronoScope auto-detects the data type based on the presence of this column and routes to the appropriate function family.

Save CosinorPy Plots: Most CosinorPy methods (2.2–2.9) expose a checkbox labeled "Save CosinorPy plots to folder" (unchecked by default). When enabled, ChronoScope calls the corresponding CosinorPy plotting function — `cosinor.plot_df_models()` for independent data or `cosinor.plot_df_models_population()` for population data — and saves the generated diagnostic figures (model fits, residual plots, waveform visualizations) to disk. Plots are never displayed inside the GUI; they are written to a subfolder named `cosinorpy_plots/` automatically created in the same directory as the loaded data file. Files are saved in both PNG and PDF format. The Periodogram method (2.1 below) does not expose this checkbox because it always saves its output as plot files — it produces no numerical table results.

2.1. Periodogram Analysis

- Purpose: Exploratory period detection prior to fitting. Generates a periodogram plot for each selected variable–condition combination to visually identify candidate oscillation periods.
- Mechanism: For each variable–condition pair, the engine loads the data in CosinorPy's internal format and calls `cosinor.periodogram(save_folder=plot_folder)`. This function internally computes the power spectrum using the method selected by `per_type` and saves the resulting figure to the `cosinorpy_plots/` output folder. No numerical results are returned to the results table; the only output is the saved plot files.
- Key Parameters: None. There are no user-adjustable parameters for this method. The period sweep range and resolution are determined internally by CosinorPy's default settings.
- Output: PNG and PDF figures saved to `cosinorpy_plots/`. Each file is named `{variable}_{condition}.png/.pdf`. No tabular output is produced.

2.2. Cosinor (independent data)

- Purpose: Fitting a cosinor model to independent (cross-sectional) data to estimate MESOR, amplitude, and acrophase.
- Mechanism: The exact CosinorPy functions called depend on the combination of `n_components` and period settings chosen by the user:

n_comp	Period	Functions called	Output
[1]	Single	<code>cosinor1.fit_group(df, period=[P])</code>	1 row. No best_model flag.
[1]	Range	<code>cosinor1.fit_group(df, period=[P1,P2,...])</code>	N rows; best period = min p-value.
[N>1] (single, N > 1)	Single	<code>cosinor.fit_group(n_components=[N], period=[P])</code> → <code>cosinor.analyse_best_models(analysis=method)</code>	1 row with extended CIs.
[N>1] (single, N > 1)	Range	<code>cosinor.fit_group(n_components=[N], period=[P1,P2,...])</code> → <code>get_best_fits(criterium)</code> → <code>filter to best period</code> → <code>analyse_best_models(...)</code>	1 row (best period) with extended CIs.
[1,2,...]	Single	<code>cosinor.fit_group(n_components=[1,2,...], period=[P])</code> → <code>get_best_models()</code> (F-test) → <code>analyse_best_models(...)</code>	M rows (one per <code>n_component</code>); best model flagged.
[1,2,...]	Range	Step 1: <code>cosinor.fit_group(n_components=[1], period=[P1...])</code> + <code>get_best_fits(criterium)</code> . Step 2: <code>cosinor.fit_group(n_components=[1,2,...], period=[best_P])</code> + <code>get_best_models()</code> + <code>analyse_best_models(...)</code>	M rows for best period; best model flagged.

- Key Parameters:
 - Period: Single value or range (min/max/step). A range triggers period optimization.
 - `n_components`: Number of harmonics to fit (1 = standard cosinor; >1 = multi-component).
 - Model Type: Specifies the statistical distribution used for regression. Three options are available:
 - Normal (default): Standard ordinary least squares cosinor regression. Suitable for continuous, approximately normally distributed data.
 - Poisson: Generalized linear model with Poisson distribution. Appropriate for raw count data.
 - Negative Binomial: Generalized linear model with negative binomial distribution.

- Best Fit Criterion: Criterion used by `get_best_fits()` to select the best period when a range is tested. Options: RSS (Residual Sum of Squares), AIC, BIC, log-likelihood.
- Analysis Method: Method used by `analyse_best_models()` to compute confidence intervals: CI (analytical) or bootstrap.
- Bootstrap Size: Number of bootstrap resamples (only active when Analysis Method = bootstrap).
- Output Interpretation (Results Table):
 - Note: Column availability varies by configuration.
 - Core Model Parameters
 - period: The period (in hours) used for this model fit. When a range is tested, each row corresponds to one tested period (`n_comp=[1]`) or the single best-selected period (multi-component paths).
 - n_components: Number of harmonic terms fitted. `n=1` is standard cosinor; `n=2` adds a 12-hour harmonic; `n=3` adds an 8-hour harmonic, etc.
 - mesor: Midline-Estimating Statistic Of Rhythm. The mean level of the fitted curve, equivalent to the vertical center of the oscillation.
 - `n_components=[1]` path: computed as the arithmetic mean of the raw data (cosinor1 does not output MESOR directly).
 - Multi-component path: taken directly from the CosinorPy fit.
 - amplitude: Half the peak-to-trough range of the fitted curve. For multi-component models this is the global amplitude of the full harmonic sum, not the amplitude of the first harmonic alone.
 - acrophase_rad / acrophase_hours: Time of the fitted curve's highest point. `acrophase_rad`: phase angle in radians (CosinorPy convention: negative values, range $[-2\pi, 0]$). `acrophase_hours`: converted to clock time (hours), using $\text{hours} = -\text{period} \times \text{acrophase_rad} / (2\pi)$. More interpretable for reporting.
 - Availability: always present when the analysis succeeds.
 - Rhythm Detection (Global F-test)
 - p_value: Overall rhythm detection p-value (Cornelissen F-test). Tests the null hypothesis that all harmonic amplitudes are zero (i.e., no rhythm exists). Low p-value → statistically significant rhythm. Available in ALL paths.
 - q_value: FDR-corrected p_value (Benjamini-Hochberg) across all tests in the group. Available in ALL paths.
 - Lack-of-Fit Test
 - p_reject / q_reject: F-test for lack of fit (Cornelissen eqs. 14–15). Tests whether the cosinor model structure is adequate for the data. Low `p_reject` (e.g., < 0.05): significant lack of fit — the cosinor curve does NOT adequately describe the data's variance structure (model should be reconsidered). High `p_reject` (> 0.05): no evidence of lack of fit — model is acceptable. `q_reject` is the FDR-corrected version.

NOT available for `n_components=[1]` path (cosinor1 does not compute it). Available in multi-component paths only.

Goodness-of-Fit Metrics

- `rss` (Residual Sum of Squares): Total squared deviation of observed values from fitted values. Lower is better. Used as the default criterium for period/model selection. NOT available for `n_components=[1]` path (cosinor1 output).
- `r_squared` / `r_squared_adj`: Proportion of variance explained by the model (R^2) and its adjusted version penalizing for model complexity. Higher is better. NOT available for `n_components=[1]` path.
- `log_likelihood`: Log-likelihood of the fitted model. Higher is better. Used as an alternative criterium for period selection. NOT available for `n_components=[1]` path.
- `aic` / `bic`: Akaike / Bayesian Information Criterion. Computed from the log-likelihood returned by CosinorPy using standard formulas: $AIC = 2k - 2 \cdot \log_likelihood$ and $BIC = k \cdot \ln(N) - 2 \cdot \log_likelihood$ where $k = 2 \times n_components + 1$ (number of model parameters: 2 per harmonic + MESOR) and N = total number of observations.
Lower values indicate a better-fitting model with less overfitting penalty. Both can be used as "Best Fit Criterium" for period selection when a range is tested. AIC penalizes model complexity more lightly than BIC; BIC is more conservative (favors simpler models) especially with larger N . NOT available for the `n_components=[1]` path (cosinor1 does not output log-likelihood).
- `resid_se` (Residual Standard Error): $\sqrt{(RSS / \text{degrees_of_freedom})}$. Average distance of observed values from the fitted curve, in the same units as the data. Lower indicates a tighter fit. NOT available for `n_components=[1]` path.
- `me` (Margin of Error): 95% margin of error for a single prediction: $ME = t_{0.975} \times \text{resid_se}$. Represents the half-width of the prediction interval. Useful to report alongside parameter estimates. NOT available for `n_components=[1]` path.

Parameter-Level Tests (from analyse_best_models)

- `p_amplitude` / `q_amplitude`: p-value (and FDR-corrected q) testing whether the amplitude is significantly different from zero. `n_components=[1]` path: available (Wald test from cosinor1). Multi-component path: available ONLY for the best model row.
- `p_acrophase` / `q_acrophase`: p-value (and q) for the acrophase estimate. Same availability as `p_amplitude`.
- `p_mesor` / `q_mesor`: p-value (and q) for the MESOR estimate. NOT available for `n_components=[1]` path (cosinor1 does not test MESOR). Available only for the best model row in multi-component paths.

Confidence Intervals

- amplitude_ci / acrophase_ci / mesor_ci: 95% confidence intervals as (lower, upper) tuples. n_components=[1] path: amplitude_ci and acrophase_ci available (from cosinor1); mesor_ci NOT available. Multi-component path: all three available ONLY for the best model row. Other rows (non-best) will have these as empty.

Peaks and Troughs

- peak_times / heights: Time(s) within one period where the fitted curve exceeds the MESOR, detected as local maxima. For single-component cosinor this is simply the acrophase. For multi-component models ($n > 1$) there can be multiple peaks within one period. NOT available for n_components=[1] path.
- trough_times: Time(s) within one period where the fitted curve falls below MESOR, detected as local minima. NOT available for n_components=[1] path.

Model Selection Flag

- best_model: Diagnostic flag indicating which row is the selected model when multiple periods or components were tested:
 - "Yes (min p-value)": best period for n_components=[1] range search.
 - "Best model (min p-value)": best n_components selected by F-test (get_best_models), when multiple components were tested.
 - "Best fit (min RSS)" / "Best fit (max log-likelihood)" / "Best fit (min AIC)" / "Best fit (min BIC)": best period selected by the chosen criterium (get_best_fits), when only a single fixed n_component was tested over a range of periods.
 - "No": valid fit but not selected as optimal.
 - None / empty: only one model was fitted (no selection needed).

2.3. Cosinor (dependent/population data)

- Purpose: Fitting a population-mean cosinor model to repeated-measures data, explicitly accounting for inter-subject variability.
- Mechanism: Routing logic mirrors the independent case, but uses the population-level function family:

n_comp	Period	Functions called	Output
[1]	Single	cosinor1.population_fit_group(df, period=P)	1 row. Amplitude, acrophase, MESOR with CIs and per-parameter p-values (analytical). No RSS, ME, resid_SE, or p_reject.
[1]	Range	Iterates over each P: cosinor1.population_fit_group(df, period=P);	N rows (one per period tested). Best period marked by minimum p-value. Same statistics

		results concatenated.	as above per row.
[N>1] (single, N > 1)	Single	cosinor.population_fit_group(df, n_components=[N], period=[P]) → get_best_models_population(criterium) → analyse_best_models_population(params_CI_analysis=method)	1 row. Full statistics including RSS, ME, resid_SE, p_reject. CIs and per-parameter p-values refined by analyse_best_models_population.
[N>1] (single, N > 1)	Range	Iterates over each P: cosinor.population_fit_group(n_components=[N], period=[P]); results concatenated → get_best_models_population(criterium) → analyse_best_models_population for best only.	N rows (one per period). Best period marked by selected criterium. All rows include basic CIs from base fit; only best row receives refined CIs from analyse_best_models_population.
[1,2,...]	Single	cosinor.population_fit_group(df, n_components=[1,2,...], period=[P]) → get_best_models_population(criterium) → analyse_best_models_population for best only.	M rows (one per n_components value). Best model marked by F-test and selected criterium. All rows include basic CIs; only best row receives refined CIs.
[1,2,...]	Range	Iterates over each P: cosinor.population_fit_group(n_components=[1,2,...], period=[P]); results concatenated → get_best_models_population(criterium) → analyse_best_models_population for best only.	N × M rows (all period × n_components combinations). Best combination marked globally by F-test and selected criterium. Only best row receives refined CIs.

- Key Parameters: Same as Cosinor (Independent), with two addition:
 - CI Analysis Method: Parameter for analyse_best_models_population(). Options: sampling (Latin Hypercube Sampling; default) or bootstrap
 - Best Fit Criterium: Only RSS is applicable for dependent data. population_fit_group does not output log-likelihood, so AIC, BIC, and log-likelihood are unavailable and will fall back to RSS.

2.4. Compare conditions (independent data)

- Purpose: Statistical comparison of rhythm parameters (MESOR, amplitude, acrophase) between two or more experimental conditions in independent samples (different subjects in each condition). All pairwise combinations are tested automatically.
- Mechanism: The library function called depends on *n_components* and the user-selected comparison strategy:
Single component (*n_components* = [1]):
 - Pooled Model (Comparison Type = "Pooled Model"): Calls *cosinor1.test_cosinor_pairs(df, pairs, period, folder)*. Fits a single joint model containing both conditions with a group-interaction term. Assumes common residual variance across groups. Compares amplitude and acrophase (but not MESOR) via independent t-tests on the difference of each parameter, using the pooled standard error. Yields higher statistical power when between-group variance is

homogeneous. Recommended when conditions are expected to differ only in rhythm shape, not in overall variance structure.

- Independent Models (Comparison Type = "Independent Models"): Calls `cosinor1.test_cosinor_pairs_independent(df, pairs, period, period2)`. Fits one cosinor model per condition independently. Compares amplitude and acrophase via t-tests using confidence-interval-derived standard errors from each separate model (no shared-variance assumption). Allows different period values per condition: when exactly two conditions are present, individual period spinboxes appear in the UI; when more than two conditions are present, the shared period value is used for all pairs. Recommended when conditions may differ substantially in variance or period.

Note: Neither single-component method provides a direct MESOR comparison. MESOR values displayed in the table are estimated from separate single-component fits run after the comparison (enrichment step) and are provided for visualization purposes only.

Multiple components (n_components = [2], [1,2,3], etc.):

- Independent method (Comparison Method = "Independent"): Calls `cosinor.compare_pairs(df, pairs, n_components=N, period=P, analysis='CI'|'bootstrap', parameters_to_analyse=[...], lin_comp=False|True)`. Fits the two conditions separately and compares the selected rhythm parameters using either analytical confidence intervals (CI) or bootstrap resampling (bootstrap). Returns FDR-corrected q-values (Benjamini-Hochberg) for each parameter and for the per-condition rhythmicity tests.
- LimoRhyde method (Comparison Method = "LimoRhyde"): Calls `cosinor.compare_pairs_limo(df, pairs, n_components=N, period=P, analysis="'|CI1'|'bootstrap1'|'CI2'|'bootstrap2')`. Uses a linear model framework derived from the limma approach (Singer et al., 2019). Fits a joint model with interaction terms and performs:
 - A global test of rhythmicity differences (p, q).
 - A joint test of all rhythm-parameter differences (p params, q params).
 - An extra-sum-of-squares F-test comparing the nested models (p(F test), q(F test)).

When analysis is non-empty (CI1, CI2, Bootstrap1, Bootstrap2), per-parameter differences and their confidence intervals are also computed and FDR-corrected.

Note: For multi-component methods, per-condition amplitude, acrophase, and MESOR displayed in the table are estimated from separate single-component re-fits (enrichment step) and serve as approximate reference values for visualization.

- Key Parameters:

- Period: Single value or range (min–max with step). When a range is specified, all period values are tested.
- n_components: Comma-separated list of harmonic counts (e.g., 1, 1,2,3). All values are tested.
- Comparison Type (n=1 only): Pooled Model: joint model, equal variance assumed. Independent Models: separate fits, unequal variance allowed.
- Comparison Method (n>1): Independent: separate model fits compared via CI or bootstrap. LimoRhyde: joint linear model with F-tests.
- Analysis Method (n>1): For Independent method: CI (faster) or Bootstrap (more robust). For LimoRhyde: None (global tests only), CI1, CI2, Bootstrap1, Bootstrap2 (additionally compute per-parameter differences).
- Bootstrap Size (n > 1, Bootstrap selected): Number of resampling iterations (default: 1000). Higher values increase precision at the cost of computation time.
- Parameters to Compare (n > 1, Independent method): Subset of {amplitude, acrophase, MESOR} to include in the comparison.
- Include Linear Component (n > 1, Independent method): If checked, a linear trend term is added and its between-condition difference is also tested.
- Output Interpretation (Results Table):
 - Period (h): Period value used for the comparison.
 - Per-condition rhythmicity (single-component Independent Models and multi-component Independent).
 - p-Cond1 / q-Cond1 F-test: p-value (and BH-corrected q-value) for rhythmicity in condition 1
 - p-Cond2 / q-Cond2: F-test p-value (and BH-corrected q-value) for rhythmicity in condition 2

These columns are not shown for the Pooled Model (which reports a single joint p-value not stored in the table) or for LimoRhyde (which uses a different model structure).
 - Per-condition rhythm parameters.
 - Amp-1 / Amp-2: Amplitude estimate for each condition
 - Acro-1 / Acro-2: Acrophase estimate for each condition (in radians; negative values indicate delay relative to time zero)
 - MESOR-1 / MESOR-2: MESOR estimate for each condition

For single-component methods: amplitude and acrophase come directly from the comparison model. MESOR is estimated from a separate individual fit. For multi-component methods: all three parameters are estimated from separate single-component re-fits for display purposes.
 - Between-condition differences
 - Amp-Diff: Amplitude difference (Cond2 – Cond1)

- p-Amp / q-Amp: p-value and BH-corrected q-value for the amplitude difference
 - CI-Amp: 95% confidence interval of the amplitude difference
 - Acro-Diff: Acrophase difference (radians; projected to $[-\pi, \pi]$)
 - p-Acro / q-Acro: p-value and BH-corrected q-value for the acrophase difference
 - CI-Acro: 95% confidence interval of the acrophase difference (radians)
 - MESOR-Diff MESOR: difference (multi-component Independent only)
 - p-MESOR / q-MESOR: p-value and BH-corrected q-value for the MESOR difference
 - CI-MESOR: 95% confidence interval of the MESOR difference
- Significance threshold: Cells with $p < 0.05$ are highlighted in green. It is recommended to interpret q-values (FDR-corrected) as the primary criterion for significance when testing multiple pairs or multiple parameters.

2.5. Compare conditions (dependent/population data)

- Purpose: Statistical comparison of rhythm parameters between two or more experimental conditions when the same subjects are measured across conditions (repeated-measures / longitudinal design). Each subject contributes one time-series per condition; the population cosinor framework estimates group-level parameters and their variability across subjects. All pairwise combinations are tested automatically.
- Mechanism:
 - Single component (`n_components = [1]`): Calls `cosinor1.population_test_cosinor_pairs(df, pairs, period, save_folder)`. Each condition is fitted independently using `population_fit_cosinor`, which estimates group-level MESOR, amplitude, and acrophase as the mean of individual-subject cosinor fits. Between-condition differences are then tested using F-tests derived from the population cosinor framework (Nelson et al., 1979):
 - Amplitude difference (F_A): F-statistic testing whether the mean amplitude differs between conditions.
 - Acrophase difference (F_{FI}): F-statistic testing whether the mean acrophase differs. A library warning is issued when acrophases differ significantly (F_{FI} : $p < 0.05$), indicating that the amplitude difference test (F_A) may not be reliable under this condition.

Note: This method tests amplitude and acrophase only. MESOR is not compared statistically. Per-condition MESOR, amplitude, and

acrophase shown in the table are estimated from separate population_fit_cosinor re-fits run after the comparison.

- Multiple components (n_components = [2], [1,2,3], etc.): Calls *cosinor.compare_pairs_population(df, pairs, n_components=N, period=P, analysis='CI'|'permutation', parameters_to_analyse=[...], lin_comp=False|True)*. Fits the population model for each condition using population_fit, then compares the selected rhythm parameters. Iterates over every (period, n_components) combination specified.
 - CI analysis: Estimates confidence intervals for between-condition parameter differences using the population cosinor CI framework. Returns 95% CIs and p-values for each selected parameter. BH-FDR-corrected q-values are computed across all pairs.
 - Permutation analysis: Uses a permutation/randomisation test. Returns differences and p-values. BH-FDR-corrected q-values are computed from the permutation p-values. No confidence intervals are produced.
- Key Parameters:
 - Period: Single value or range (min–max with step). All values tested.
 - n_components: Comma-separated harmonic counts. All values tested.
 - Analysis Method (n > 1 only): CI: confidence-interval-based comparison (returns CIs). Permutation: randomisation test (no CIs).
 - Parameters to Compare (n > 1 only): Subset of {amplitude, acrophase, MESOR} to include in the comparison.
 - Include Linear Component (n > 1 only): If checked, adds a linear trend term and tests its between-condition difference.
- Output Interpretation (Results Table):
 - Period (h): Period value used for the comparison.
 - Per-condition rhythmicity:
 - p-Cond1 / p-Cond2: p-value for rhythmicity detection in each condition.
For multi-component: from the population_fit F-test; BH-corrected q-values (q-Cond1, q-Cond2) are also provided.
For single-component: from the population_fit_cosinor F-test (same re-fit used for per-condition parameters); q-Cond1 / q-Cond2 are N/A.
 - Per-condition rhythm parameters (enrichment re-fit — all methods):
 - Amp-1 / Amp-2: Mean amplitude per condition (from population_fit_cosinor re-fit)
 - Acro-1 / Acro-2: Mean acrophase per condition in radians (from re-fit)
 - MESOR-1 / MESOR-2: Mean MESOR per condition (from re-fit)
Note: These values always come from single-component (n=1) population_fit_cosinor re-fits performed after the comparison,

regardless of the `n_components` selected. They are provided for reference and visualization; the statistical comparison uses the appropriate model (single- or multi-component) internally.

- Between-condition differences:
 - Amp-Diff: Amplitude difference (Cond2 – Cond1)
 - p-Amp / q-Amp: p-value and BH-corrected q-value for amplitude difference.
 - CI-Amp ($n > 1$, CI analysis only): 95% CI of the amplitude difference.
 - Acro-Diff: Acrophase difference (radians; projected to $[-\pi, \pi]$)
 - p-Acro / q-Acro: p-value and BH-corrected q-value for acrophase difference
 - CI-Acro ($n > 1$, CI analysis only): 95% CI of the acrophase difference
 - MESOR-Diff ($n > 1$ only): MESOR difference
 - p-MESOR / q-MESOR ($n > 1$ only): p-value and BH-corrected q-value for MESOR difference
 - CI-MESOR ($n > 1$, CI analysis only): 95% CI of the MESOR difference

2.6. Nonlinear cosinor (independent data)

- Purpose: Detecting rhythms with non-stationary characteristics, specifically: damped oscillations (amplitude decreasing over time), forced oscillations (amplitude increasing over time), and signals with a systematic linear trend.
- Model: $y = A + B \cdot \exp(C \cdot t) \cdot \cos(2\pi t/P + \phi) + D \cdot t$, where C is the amplification coefficient (negative = damped, positive = forced) and D is the linear trend component.
- Mechanism: Routes to `cosinor_nonlin` functions based on parameter settings:

n_comp	Period	Functions called
[1]	Single period	<code>cosinor_nonlin.fit_generalized_cosinor_group(df, period=P)</code> – CIs calculated analytically via <code>curve_fit</code> . No bootstrap needed.
[1]	Range of periods	Loops over each P : <code>fit_generalized_cosinor_group(df, period=P)</code> per period; best by minimum p-value.
[N>1]	Single period	<code>fit_generalized_cosinor_n_comp_group(df, period=P, n_components=N)</code> → <code>bootstrap_generalized_cosinor_n_comp_group(df, period=P, n_components=N, bootstrap_size=B)</code> — bootstrap always applied to obtain CIs for amplitude and acrophase.
[N>1]	Range of periods	Loops over each P : fit + bootstrap per period; best model selected by minimum p-value.
[1,2,...]	Single period	<code>fit_generalized_cosinor_n_comp_group_best(df, period=P,</code>

		n_components=[1,2,...]) (selects best n by p-value) → bootstrap_generalized_cosinor_n_comp_group_best(df, df_best_models, bootstrap_size=B) — bootstrap always applied.
[1,2,...]	Range of periods	Loops over each P: fit_best + bootstrap_best per period; best period selected by minimum p-value.

- Key Parameters:

- Period: Single value or range.
- n_components: Number of harmonics.
- Bootstrap Size: Number of bootstrap resamples for CI estimation of amplitude and acrophase (applied when n_components > 1; not needed for single-component fits, which use analytical statistics).

- Output Interpretation (Results Table):

Note: Column availability varies by configuration.

Core Model Parameters

- period: The period (in hours) used for this model fit. When a range is tested, each row corresponds to one tested period.
- n_components: Number of harmonic terms in the nonlinear model. In the [1, 2, ...] path, this reflects the number of harmonics automatically selected by CosinorPy (best by p-value), not the full list provided as input.
- mesor: Midline-Estimating Statistic Of Rhythm — the constant A in the model, representing the mean level of the oscillation at time zero.
n_components=[1] path: NOT available (N/A).
fit_generalized_cosinor_group does not expose the MESOR parameter A in its output. - Multi-component paths ([N>1] and [1,2,...]): taken directly from the CosinorPy fit (fit_generalized_cosinor_n_comp_group or fit_generalized_cosinor_n_comp_group_best).
- amplitude: Half the peak-to-trough range of the fitted oscillatory component (parameter B). For multi-component models, this is the global amplitude of the full harmonic sum, not the amplitude of the first harmonic alone.
- acrophase_rad / acrophase_hours: Phase of the fitted oscillation.
acrophase_rad: phase angle in radians (CosinorPy convention: range $[-\pi, \pi]$). acrophase_hours: converted to clock time using $\text{hours} = -\text{period} \times \text{acrophase_rad} / (2\pi)$.

Rhythm Detection

- p_value: Overall rhythm detection p-value (Cornelissen F-test). Tests the null hypothesis of no rhythm (all harmonic amplitudes = 0). Low p-value → statistically significant rhythm. Available in all configurations.
- q_value: FDR-corrected p_value (Benjamini-Hochberg) across all tests in the analysis. Available in all configurations.

Nonlinear-Specific Parameters (unique to this method)

- amplification (C): Estimated exponential amplification coefficient. Governs how the oscillation's amplitude changes over time.
 - $C < 0$: damped oscillation (amplitude decreases over time)
 - $C = 0$: sustained oscillation (standard cosinor)
 - $C > 0$: forced oscillation (amplitude increases over time)
- p_amplification / q_amplification: p-value (and FDR-corrected q) testing whether C is significantly different from zero (i.e., whether the damping/forcing is statistically significant).
- amplification_ci: 95% confidence interval for C as (lower, upper).
 - n_components=[1] path: computed analytically from the curve_fit covariance matrix.
 - Multi-component paths: taken from the bootstrap or the fit result.
- lin_comp (D): Estimated linear trend component. Represents a systematic increase or decrease in the signal's baseline over time (independent of the oscillation).
- p_lin_comp / q_lin_comp: p-value (and FDR-corrected q) testing whether D is significantly different from zero.
- lin_comp_ci: 95% confidence interval for D as (lower, upper).

Parameter-Level Tests for Amplitude and Acrophase

- p_amplitude / q_amplitude: p-value (and FDR-corrected q) testing whether the amplitude B is significantly different from zero.
 - n_components=[1] path: computed analytically from curve_fit (Wald test).
 - Multi-component paths: from bootstrap resampling.
- p_acrophase / q_acrophase: p-value (and q) for the acrophase estimate.

Confidence Intervals

- amplitude_ci / acrophase_ci: 95% confidence intervals as (lower, upper) tuples.
 - n_components=[1] path: computed analytically from the curve_fit covariance matrix (no bootstrap needed).
 - Multi-component paths: computed by bootstrap resampling (bootstrap_generalized_cosinor_n_comp_group or *_best).

Model Selection Flag

- best_model: Diagnostic flag indicating the selected model when multiple periods were tested. Only relevant when a period range is specified.
 - "Yes (min p-value)": this period produced the lowest overall p-value and is the selected fit.
 - "No": valid fit for this period but not selected as optimal.
 - None / empty: only one period was tested (no selection needed).
- Note: When n_components=[1,2,...] (auto-select path), the best number of harmonics is selected internally by CosinorPy (get_best_model based on F-test p-value) before ChronoScope receives the result. The

best_model flag applies only to period selection across a range, not to component selection.

2.7. Nonlinear cosinor (dependent/population data)

- Purpose: Detect rhythms with non-stationary characteristics — damped oscillations (amplitude decreasing over time), forced oscillations (amplitude increasing over time), and signals with a systematic linear trend — in repeated-measures designs. Use this method when the same subjects (animals, cell cultures, biological replicates) are measured across the entire time course, so that individual-level parameter estimates can serve as the sampling distribution for population-level inference.
- Model: $y = A + B \cdot \exp(C \cdot t) \cdot \cos(2\pi t/P + \phi) + D \cdot t$, where C is the amplification coefficient (negative = damped, positive = forced) and D is the linear trend component. The model is identical to Nonlinear (Independent); the difference lies in the statistical inference strategy.
- Statistical Mechanism: The population-level parameters and their standard errors are derived from the variability between replicates, not from the curvature of the likelihood surface. For each replicate, the nonlinear model is fitted individually, yielding a set of per-replicate parameter estimates. The population mean, standard deviation, t-statistic, and confidence intervals are then computed across those per-replicate estimates (using t-distribution with $k-1$ degrees of freedom, where k = number of replicates). This approach is analogous to a one-sample t-test on the replicate parameter distributions. As a consequence, bootstrap is neither needed nor available for this method.
- Mechanism: Uses the *population_* variants of the same nonlinear functions.

n_comp	Period	Functions called
[1]	Single period	cosinor_nonlin.population_fit_generalized_cosinor_group(df, period=P) — CIs calculated analytically from inter-replicate variance.
[1]	Range of periods	Loops over each P: population_fit_generalized_cosinor_group(df, period=P) per period; best period by minimum p-value.
[N>1]	Single period	cosinor_nonlin.population_fit_generalized_cosinor_n_comp_group(df, period=P, n_components=N)
[N>1]	Range of periods	Loops over each P: population_fit_generalized_cosinor_n_comp_group(...) per period.
[1,2,...]	Single period	cosinor_nonlin.population_fit_generalized_cosinor_n_comp_group_best(df, period=P, n_components=[1,2,...])
[1,2,...]	Range of periods	Loops over each P: population_fit_generalized_cosinor_n_comp_group_best(...) per period.

- Key Parameters: Same as Nonlinear (Independent), excluding Bootstrap Size (not applicable to population data).
- Output Interpretation (Results Table)

- Note: Column availability varies by configuration.

Core Model Parameters

- period: The period (in hours) used for this model fit. When a range is tested, each row corresponds to one tested period.
- mesor: The constant A in the model, representing the mean level of the oscillation (Midline-Estimating Statistic Of Rhythm).
n_components=[1]: population_fit_generalized_cosinor_group does not expose parameter A in its output DataFrame. This is a library-level constraint, identical to the behaviour in Nonlinear (Independent).
Multi-component paths ([N>1] and [1,2,...]): taken directly from rhythm_params — the mean population curve is evaluated over one full period to extract amplitude, acrophase, and mesor from the fitted waveform.
- amplitude: Half the peak-to-trough range of the fitted oscillatory component (parameter B). For multi-component models, this is the global amplitude of the full harmonic sum evaluated from the population-mean fitted curve.

Rhythm Detection

- p_value: Overall rhythm detection p-value. Tests the null hypothesis of no rhythm (all harmonic amplitudes = 0). Low p-value → statistically significant rhythm. Available in all configurations.
- q_value: FDR-corrected p-value (Benjamini-Hochberg) across all tests in the analysis. Available in all configurations.

Nonlinear-Specific Parameters (unique to this method family)

- amplification (C): Estimated exponential amplification coefficient. Governs how the oscillation's amplitude changes over time.
 - $C < 0$: damped oscillation (amplitude decreases over time)
 - $C = 0$: sustained oscillation (standard cosinor behaviour)
 - $C > 0$: forced oscillation (amplitude increases over time)
- p_amplification / q_amplification: p-value (and FDR-corrected q) testing whether C is significantly different from zero (i.e., whether the damping/forcing is statistically significant)
- amplification_ci: 95% confidence interval for C as (lower, upper). Computed from inter-replicate variance (t-distribution, k-1 df).
- lin_comp (D): Estimated linear trend component. Represents a systematic increase or decrease in the signal's baseline over time, independent of the oscillation.
- p_lin_comp / q_lin_comp: p-value (and FDR-corrected q) testing whether D is significantly different from zero.
- lin_comp_ci: 95% confidence interval for D as (lower, upper).

Parameter-Level Tests for Amplitude and Acrophase

- p_amplitude / q_amplitude and amplitude_ci:
 - n_components=[1]: population_fit_generalized_cosinor_group computes the t-statistic for B from inter-replicate variance and returns p(amplitude), q(amplitude), and CI(amplitude).
 - Multi-component([N>1] and [1,2,...]): population_fit_generalized_cosinor_n_comp_group and *_best do not expose parameter-level tests for amplitude. In the independent version, bootstrap fills this gap; that option does not exist for population data.
- p_acrophase / q_acrophase and acrophase_ci:
 - n_components=[1]: Computed from inter-replicate circular variance (circular mean and standard deviation of per-replicate acrophase estimates).
 - Multi-component paths: NOT available, for the same reason as above.

Model Selection Flag

- best_model: Diagnostic flag indicating the selected model when multiple periods were tested. Only relevant when a period range is specified.
 - "Yes (min p-value)": this period produced the lowest overall p-value and is the selected fit.
 - "No": valid fit for this period but not selected as optimal.
 - None / empty: only one period was tested (no selection needed).
- Note: When n_components=[1,2,...] (auto-select path), the best number of harmonics is selected internally by CosinorPy before ChronoScope receives the result. The best_model flag applies only to period selection across a range, not to component selection.

2.8. Nonlinear compare (independent data)

- Purpose: Statistical comparison of nonlinear generalized cosinor parameters — including amplitude, acrophase, amplification, and linear trend — between two or more independent conditions. All pairwise combinations are tested automatically. Designed for datasets where observations in each condition come from different (non-paired) subjects or sampling units.
- Mechanism: Three comparison strategies are available:
 - Dependent Model (same period for both groups, 1 component): Calls cosinor_nonlin.fit_generalized_cosinor_compare_pairs_dependent(df, pairs, period=P). Fits a single combined nonlinear model to both conditions simultaneously. A binary group indicator H (H = 0 for condition 1, H = 1 for condition 2) is incorporated into the model, which takes the form: $Y(t) = (A + A_0H) + (B + B_0H) \cdot e^{((C + C_0H) \cdot t)} \cdot \cos(2\pi t/P + \phi + \phi_0H) + (D + D_0H) \cdot t$

The base parameters (A, B, C, D, ϕ) describe condition 1; the interaction terms (A_0 , B_0 , C_0 , D_0 , ϕ_0) directly represent the between-condition differences and are estimated jointly from the nonlinear least-squares fit. Both conditions are constrained to share a single period P. Standard errors for each difference term are derived analytically from the covariance matrix of the joint fit. An overall F-test p-value (global test of any rhythm difference) is additionally reported alongside per-parameter tests. This strategy offers higher statistical power when the homogeneous-period and equal-residual-variance assumptions hold. Recommended when the two conditions are expected to oscillate at the same period and when between-group variance is similar.

- Independent Model (allows different periods, 1 component): Calls `cosinor_nonlin.fit_generalized_cosinor_compare_pairs_independent(df, pairs, period1=P1, period2=P2)`. Fits two independent generalized cosinor models, one per condition. Parameter differences $d_\theta = \theta_2 - \theta_1$ are computed after fitting, and a Welch-type t-test is applied using SEs derived from each model's individual 95% CIs: $SE(d_\theta) = \sqrt{SE^2_{\theta_1} + SE^2_{\theta_2}}$

The t-statistic is evaluated against a t-distribution with combined degrees of freedom ($n_1 + n_2 - k_1 - k_2$), where k_1 and k_2 are the number of estimated parameters per model. Acrophase differences are projected to $[-\pi, \pi]$ before testing. Conditions may have different periods P1 and P2. This strategy is more robust than the joint model when between-group variance differs substantially or when the two conditions are suspected to oscillate with different intrinsic periods (e.g., short-photoperiod vs. long-photoperiod). When three or more conditions are present, per-condition period entry is not supported and the shared period is used for all pairs.

- Multi-component with Bootstrap ($n_components > 1$): Calls `cosinor_nonlin.compare_pairs_n_comp_bootstrap_group(df, pairs, n_components=N, period=P, bootstrap_size=B)`. Extends the generalized cosinor to N harmonic components. For each pair:
 - 1. A joint multi-component nonlinear fit is run per condition via `compare_pairs_n_comp_basic` to obtain point estimates and analytical CIs for all parameters.
 - 2. Amplitude and acrophase: inference uses non-parametric bootstrap resampling (B iterations). The bootstrap SE and t-test yield p-values and CIs that are robust to non-normality of residuals.
 - 3. Amplification (C) and linear component (D): inference uses the analytical SE from the joint model fit. Bootstrap is not applied to these parameters.

FDR correction (Benjamini-Hochberg) is applied to all pairs jointly for each parameter. Only a single shared period P is supported for this strategy; per-condition periods are not available. When n_components contains multiple values (e.g., 1, 2), Strategy 1 or 2 is applied for N = 1 and Strategy 3 for each N > 1, with all results accumulated in the table.

- Key Parameters:
 - Use dependent model: Controls period strategy. Unchecked (default): allows different periods per condition. Checked: enforces shared period. Has no effect when n_components > 1 (Strategy 3 always uses a single shared period).
 - Components: Number of harmonic components. Single value or comma-separated list (e.g., 1, 2, 1,2). N=1 uses analytical statistics; N>1 uses bootstrap for amplitude and acrophase.
 - Bootstrap Size: Bootstrap iterations. Visible only when at least one component > 1 is specified. Determines the precision of amplitude and acrophase p-values. Recommended minimum: 500 for exploratory analysis, 1000–2000 for publication.
- Output Interpretation (Results Table)
 - Period (h): Period used for the comparison (period1 of the pair; when period1 ≠ period2, period2 is not separately shown in this column)
 - Per-condition rhythm parameters
 - Amp-1 / Amp-2: Amplitude estimate per condition
 - Acro-1 / Acro-2: Acrophase estimate per condition (radians)
 - MESOR-1 / MESOR-2: MESOR estimate per condition
 - Note: These values are estimated from auxiliary linear cosinor fits (1 component, cosinor1.fit_cosinor) run after the nonlinear comparison (enrichment step). They are provided for visualization and approximate reference only. They do not come from the nonlinear model and should not be reported as the primary parameter estimates for nonlinear fits. For nonlinear per-condition estimates, use Method 6 (Nonlinear Analysis, Independent) or Method 7 (Nonlinear Analysis, Dependent) on each condition individually.
 - Between-condition differences
 - Amp-Diff: $d_amplitude = Amplitude(Cond2) - Amplitude(Cond1)$
 - p-Amp: p-value for $H_0: d_amplitude = 0$ (t-test for N=1; bootstrap t-test for N>1)
 - q-Amp: BH-corrected p-value across all pairs
 - CI-Amp: 95% confidence interval of d_amplitude
 - Acro-Diff: $d_acrophase$ (radians, projected to $[-\pi, \pi]$) = $Acrophase(Cond2) - Acrophase(Cond1)$
 - p-Acro: p-value for $H_0: d_acrophase = 0$
 - q-Acro: BH-corrected p-value
 - CI-Acro: 95% confidence interval of d_acrophase (radians)

- Amplif-Diff: $d_amplification = C(Cond2) - C(Cond1)$. Positive: Cond2 rhythm amplifies faster (or damps less) than Cond1
- p-Amplif: p-value for $H_0: d_amplification = 0$ (always analytical, not bootstrapped)
- q-Amplif: BH-corrected p-value
- CI-Amplif: 95% confidence interval of $d_amplification$
- LinComp-Diff: $d_lin_comp = D(Cond2) - D(Cond1)$. Positive: Cond2 has a steeper upward linear drift
- p-LinComp: p-value for $H_0: d_lin_comp = 0$ (always analytical)
- q-LinComp: BH-corrected p-value
- CI-LinComp: 95% confidence interval of d_lin_comp

Note: The MESOR values shown under per-condition parameters come from the auxiliary linear fit and are not subjected to formal comparison here.

Significance threshold and multiple comparison correction

All p-values within each parameter (amplitude, acrophase, amplification, lin_comp) are corrected for multiple comparisons across all pairs using the Benjamini-Hochberg FDR procedure, yielding q-values. Cells with $p < 0.05$ are highlighted in green. It is recommended to use q-values as the primary criterion for significance when testing more than two conditions. When interpreting results, note that amplitude and acrophase share a mathematical relationship via the circular nature of the phase: a rhythm that differs in acrophase without a change in amplitude, or vice versa, should still be interpreted as a meaningfully different rhythm.

2.9. Nonlinear compare (dependent/population data)

- Purpose: This method tests whether two or more conditions differ in their nonlinear rhythm parameters — amplitude, acrophase, exponential amplification, and linear trend — in repeated-measures designs, where the same biological units (animals, cell cultures, or biological replicates) are observed across the full time course. Use this method when data have a *subject* column identifying each replicate, and when you suspect that the oscillation's amplitude is not stationary (i.e., the signal is damped or forced) or that a systematic linear drift is present in the baseline. The inferential advantage of the population design over the independent design is that inter-replicate variability provides the sampling distribution directly, without requiring bootstrap resampling.
- Model: Each replicate is fitted individually with the generalized cosinor model:

$$y = A + B \cdot \exp(C \cdot t) \cdot \cos(2\pi t/P + \phi) + D \cdot t$$

A: MESOR: time-averaged mean level of the signal
 B: Amplitude: maximum displacement from the MESOR at $t = 0$

C: Amplification coefficient: governs exponential change in oscillatory amplitude over time. $C < 0$ produces a damped oscillation (envelope decaying over time), $C = 0$ recovers the standard sustained cosinor, and $C > 0$ produces a forced oscillation (envelope growing over time)

D: Linear component: systematic linear drift in the baseline

P: Period (hours), fixed by the user

ϕ : Acrophase: timing of the oscillatory peak (radians)

- Mechanism: Two strategies available:
 - Single component ($n_components = [1]$): calls `cosinor_nonlin.population_fit_generalized_cosinor_compare_pairs(df, pairs, period1=P1, period2=P2)`. Allows different periods per condition.
 - Multi-component ($[N]$ where $N > 1$ or $[1, 2, \dots]$): calls `cosinor_nonlin.population_compare_pairs_n_comp_group(df, pairs, n_components=N, period=P)`.
Note on single vs. multiple component lists: When $n_components = [1]$ exactly, the single-component function is used and full statistics (p-values, q-values, CIs) are available for all four parameters. When $n_components$ contains any list with more than one element — including $[1, 2, 3]$ — the multi-component function is used for every value in the list. This ensures a consistent output structure across all rows. The cost is that `d_amplitude` and `d_acrophase` lose their statistical annotation for $n_comp = 1$ rows within a multi-value list.
- Key Parameters:
 - Period: The period (in hours) assumed for both conditions. A single value is required; unlike the fit methods, compare methods do not scan a range of periods. Set this to the expected or previously determined period of the rhythm under investigation.
 - Components: The number of harmonic components to include in the model. $[1]$ fits a single-frequency generalized cosinor. $[2]$ or $[3]$ accommodate more complex waveforms (e.g., non-sinusoidal oscillations with shoulders or secondary peaks). Multiple values (e.g., $[1, 2, 3]$) produce one comparison row per component count, allowing you to assess whether the conclusion about a given parameter difference is robust across model complexity. Bootstrap Size is not available for this method; statistics are derived from inter-replicate variance regardless of component count.
- Output Interpretation (Results Table)
 - Amp-1 / Amp-2: Population-mean amplitude estimate for condition 1 and condition 2, respectively. Derived from a linear cosinor fit of each condition's replicate ensemble.
 - Acro-1 / Acro-2: Population-mean acrophase (radians) for each condition.
 - MESOR-1 / MESOR-2: Population-mean MESOR for each condition.

- Amplif-1 / Amplif-2: Population-mean amplification coefficient (C) for each condition, derived from the individual nonlinear fit. Negative values indicate damped oscillations; positive values indicate forced oscillations.
- LinComp-1 / LinComp-2: Population-mean linear trend component (D) for each condition.
- p-Cond1 / p-Cond2: Rhythm detection p-value for each condition individually (from `cosinor1.population_fit_cosinor`). These test whether a significant rhythm is present in each condition separately, independent of the comparison. Useful for identifying cases where a rhythmic difference is detected even though one or both conditions are not individually significant.

Amplitude Difference

- Amp-Diff (`d_amplitude`): Difference in amplitude between condition 2 and condition 1 ($\text{Cond2} - \text{Cond1}$). A positive value means the oscillation is larger in condition 2.
- p-Amp / q-Amp: p-value and FDR-corrected q-value (Benjamini-Hochberg) testing whether the amplitude difference is significantly different from zero. Available for `n_components = [1]` only.
- CI-Amp: 95% confidence interval for the amplitude difference, reported as (lower, upper). Available for `n_components = [1]` only.

Acrophase Difference

- Acro-Diff (`d_acrophase`): Difference in acrophase between condition 2 and condition 1 (radians, projected to $[-\pi, \pi]$). A positive value indicates a phase delay (the peak occurs later) in condition 2; a negative value indicates a phase advance.
- p-Acro / q-Acro: p-value and FDR-corrected q-value for the acrophase difference. Available for `n_components = [1]`: computed using circular statistics (circular mean and standard deviation of per-replicate acrophase estimates).
- CI-Acro: 95% confidence interval for the acrophase difference (radians). Available for `n_components = [1]` only.

Amplification Difference (Nonlinear-Specific)

- Amplif-Diff (`d_amplification`): Difference in the exponential amplification coefficient C between condition 2 and condition 1. Interpretation:
 - Positive value: the oscillation's amplitude grows faster over time (or decays more slowly) in condition 2 relative to condition 1.
 - Negative value: the oscillation decays faster (or grows more slowly) in condition 2.
 - A value near zero indicates no differential change in the envelope between conditions, even if both conditions individually show non-zero C.
- p-Amplif / q-Amplif: p-value and FDR-corrected q-value testing whether the amplification difference is significantly different from zero.

- CI-Amplif: 95% confidence interval for the amplification difference, as (lower, upper). Available for all configurations.
- Linear Component Difference (Nonlinear-Specific)
- LinComp-Diff (d_lin_comp): Difference in the linear trend coefficient D between condition 2 and condition 1. A significant positive value means the baseline rises more steeply in condition 2; a significant negative value means it rises less steeply (or declines more).
- p-LinComp / q-LinComp: p-value and FDR-corrected q-value for the linear component difference.
- CI-LinComp: 95% confidence interval for the linear component difference, as (lower, upper). Available for all configurations.

3. MODULE III: CircaCompare (circacompare_analysis.py)

A wrapper for the CircaCompare method. This module uses non-linear least squares optimization to statistically model and compare rhythm parameters (MESOR, amplitude, and acrophase) between two groups.

3.1. Single Fit (Robust Fitting)

- Purpose: Determining the phase, amplitude, and MESOR of a single group with high resistance to outliers.
- Mechanism: Utilizes non-linear regression (`scipy.optimize.least_squares`) applying the model $y = \text{MESOR} + \text{Amplitude} \cdot \cos(2\pi t/T - \phi)$, where time t is internally converted to radians based on the specified period. The optimizer is initialized with random starting values and retried up to the maximum iterations limit to avoid local minima.
- Key Parameters:
 - Period Range: The assumed period of oscillation in hours (default: 24h). When a range is specified (e.g., 20–28h with a step), the model is fitted independently for each period value in the range, returning one result row per period. No automatic period selection is performed.
 - Loss Function: Determines how residuals are penalized during optimization.
 - 'linear': Standard least squares. No outlier protection.
 - 'soft_l1' / 'huber': Moderate robustness; reduce the influence of outliers without fully discarding them.
 - 'cauchy'/'arctan': Strong downweighting of extreme outliers. Use when data contains severe contamination.
 - F-Scale: Defines the residual magnitude below which data points are treated as inliers (approximately linear) versus outliers (downweighted). Lower values make the model stricter. Has no effect when using linear loss.
 - Max Iterations: The maximum number of optimization attempts with random starting values before declaring failure. Increase if the fit does not converge.
- Output Interpretation (Results Table):
 - MESOR / Amplitude / Acrophase (h): The estimated circadian parameters for the fitted model.
 - CI (MESOR) / CI (Amplitude) / CI (Acrophase, rad): Approximate 95% confidence intervals (lower, upper) for each parameter, derived from the covariance matrix of the optimization.
 - SE (MESOR) / SE (Amplitude) / SE (Acrophase): Standard errors for each parameter, estimated from the Jacobian matrix at the solution.

Note: these are approximate and assume local linearity around the optimum; they may be unreliable if the fit is near a local minimum.

3.2. Compare Groups (Differential Analysis)

- Purpose: Testing if two biological conditions or groups differ significantly in their MESOR, amplitude, or acrophase (phase).
- Mechanism: Fits a joint parametric model to both groups simultaneously:

$$y = (M + \delta M \cdot g) + (A + \delta A \cdot g) \cdot \cos(2\pi t/T - (\phi + \delta \phi \cdot g))$$
 where g is the group indicator (0 = reference, 1 = comparison), M , A , and ϕ are the baseline parameters for Group 0, and δM , δA , $\delta \phi$ are the differences (Group 1 – Group 0). The same robust optimization engine used in Single Fit is applied to the combined dataset from both conditions.
- Key Parameters:
 - Identical to Single Fit (Period, Loss Function, F-Scale, Max Iterations). The same robust optimization engine is used on the combined dataset from both conditions.
- Output Interpretation (Results Table):
 - MESOR-1 / Amp-1 / Acro-1 (h): Estimated parameters for the reference condition (Group 0 / Cond1).
 - MESOR-2 / Amp-2 / Acro-2 (h): Estimated parameters for the comparison condition (Group 1 / Cond2), derived by adding the fitted differences to the Group 0 baseline.
 - MESOR-Diff / Amp-Diff / Acro-Diff (h): The fitted difference parameters (Group 1 – Group 0) for each circadian component.
 - CI (MESOR-Diff) / CI (Amp-Diff) / CI (Acro-Diff, rad): The 95% confidence intervals for the difference parameters, estimated from the Jacobian of the joint optimization. The acrophase CI is reported in radians.
 - sig (MESOR) / sig (Amp) / sig (Acro): Significance of each difference, evaluated by whether its 95% CI excludes zero. A cell colored green indicates a significant difference (CI does not include 0); uncolored indicates no significant difference. CircaCompare uses this CI-based criterion rather than classical p-values.

4. MODULE IV: RhythmCount (rhythmcounanalysis.py)

A wrapper for the RhythmCount library (Velikajne et al. 2022). This module fits cosinor models within a generalized linear model (GLM) framework for count-valued time series — RNA-seq read counts, neuronal spike counts, locomotor activity event tallies, or any non-negative discrete data for which the Gaussian residual assumption of standard cosinor is violated. Five count distributions are supported, allowing the user to select the model that best matches the variance structure of their data.

Supported count distributions:

- Poisson: Assumes variance equals the mean (equidispersion). Appropriate for clean, low-count sequencing data.
- Generalized Poisson: Flexible dispersion parameter; accommodates both under- and over-dispersed data.
- Negative Binomial: Models overdispersion (variance > mean). The typical choice for RNA-seq data with biological replicates.
- Zero-Inflated Poisson (ZIP): Combines a Poisson process with a separate excess-zero component. Use when the data contain more zeros than a Poisson model would predict.
- Zero-Inflated Negative Binomial (ZINB): Combines overdispersion and excess zeros. Appropriate for sparse behavioral recordings or lowly-expressed transcripts.

4.1 Fit Single Model

- Purpose: Fitting a single, explicitly specified count distribution and harmonic complexity to a dataset. Recommended when the investigator has prior knowledge of the appropriate distribution, or when comparing specific model variants against a predefined baseline.
- Mechanism: Constructs a harmonic design matrix from sine and cosine regressors derived from the specified period and number of components:
$$\log(\mu_i) = \beta_0 + \sum_{k=1}^N [\beta_{2k-1} \cdot \sin(2\pi k \cdot t_i/T) + \beta_{2k} \cdot \cos(2\pi k \cdot t_i/T)]$$
The selected GLM is fitted using the corresponding statsmodels class (e.g., Poisson, NegativeBinomialP, ZeroInflatedNegativeBinomialP). Rhythmicity significance is assessed via a likelihood ratio test comparing the full harmonic model against an intercept-only null model. Amplitude, MESOR, and peak times are extracted from the dense predicted curve after fitting.
- Key Parameters:
 - Count Distribution: The statistical model for the response variable. Select based on the variance structure of your data (see guidance above). Default: Poisson.
 - N Components: Number of harmonic terms in the model. N = 1 fits a single sinusoid (fundamental frequency only). N = 2 or N = 3 adds higher harmonics for non-sinusoidal waveforms. Default: 1.
 - Period: Expected oscillation period in the same time units as the input data. Default: 24.0 (hours).

- Max Iterations / Max Function Evaluations: Upper limits for the numerical optimizer. Increase if the model fails to converge on complex or noisy data. Defaults: 5000 each.
- Clean Data: If enabled, applies per-timepoint quantile filtering (15th–85th percentile) to remove outliers before fitting. Default: disabled.
- Output Interpretation (Results Table):
 - Amplitude: Half the range of the fitted curve within one period. Approximates the cosinor amplitude for multi-component fits.
 - MESOR: Midline of the fitted curve (mean of minimum and maximum predicted values), representing the rhythm-adjusted mean count.
 - Peaks (h): Time(s) within one period where the fitted curve exceeds 75% of its maximum value. Corresponds to the acrophase for unimodal rhythms; multiple values indicate multi-modal waveforms.
 - Heights: Predicted count values at each detected peak.
 - LLR p-value: Likelihood ratio test p-value for the test of zero rhythmicity (full harmonic model vs. intercept-only null). The primary significance criterion.
 - AIC / BIC: Akaike and Bayesian information criteria. Lower values indicate better model fit, penalized for complexity. Used for model comparison across distributions or component levels.
 - Pseudo-R² (McFadden): Proportion of log-likelihood explained relative to the null model. Ranges from 0 to 1; values above 0.2 are generally considered good fits in GLM contexts.
 - RSS: Residual sum of squares between observed and fitted values. Useful for comparing goodness of fit on the original count scale.
 - Log-Likelihood: Log-likelihood of the fitted model. Used internally for AIC, BIC, and the Vuong test.

4.2 Fit All Models

- Purpose: Systematically fitting all combinations of the selected count distributions and harmonic component levels and returning a comprehensive comparison table. This method does not perform automatic selection — it presents all results for inspection so that the investigator can make an informed choice.
- Mechanism: Iterates over every (distribution × n_components) combination and calls the single-model fitting procedure for each. The result is a matrix of fitted models, one row per combination. Optionally renders a multi-panel figure with one panel per distribution, overlaying fits for each component level.
- Key Parameters:
 - Count Distributions: Subset of distributions to include in the grid. Any combination of the five supported distributions. Default: all five.

- N Components: List of harmonic complexity levels to evaluate. Example: [1, 2, 3] fits $N = 1$, $N = 2$, and $N = 3$ for each selected distribution. Default: [1, 2, 3].
 - Period, Max Iterations, Max Function Evaluations, Clean Data: As in 4.1.
 - Plot Models: If enabled, displays a comparison figure with fitted curves for all combinations. Default: disabled.
 - Output Interpretation (Results Table):
 - One row per (distribution, n_components) combination. Total rows = number of selected distributions $\times$ number of component levels (e.g., $5 \times 3 = 15$ rows). All output fields are identical to those of Fit Single Model (see 4.1), with two additional identifying columns:
 - Count Model: Name of the distribution for that row.
 - N Components: Number of harmonic terms for that row.
- Use AIC or BIC to rank rows — lower is better. For nested comparisons (same distribution, increasing N), a decrease in $AIC > 2$ units supports the more complex model.

4.3 Fit Best Model (Automatic Model Selection)

- Purpose: Automatically identifying the count distribution and harmonic complexity that best describe the data, without requiring the user to manually inspect the full comparison grid. This is the recommended entry point for hypothesis-driven analysis when the appropriate model is not known a priori.
- Mechanism: Performs a two-stage model selection procedure on the full grid from 4.2:
 - Stage 1 (default order): Selects the best number of harmonic components across distributions using nested model comparison (F-test or information criteria).
 - Stage 2: Given the selected N , chooses the best count distribution using non-nested comparison (AIC, BIC, or Vuong test).

The alternative order (Stage 1: distribution; Stage 2: components) is available via the Evaluation Order parameter.

Model comparison tests:

- AIC / BIC: Selects the model with the lower information criterion. BIC applies a stronger penalty for model complexity than AIC.
- F-test: Tests whether adding more harmonic components significantly improves fit (nested models). Selects the more complex model if F p-value < 0.05 .
- Vuong test (Vuong 1989): Non-nested likelihood ratio test using per-observation log-likelihoods. A Z-statistic p-value < 0.05 indicates that the model with higher likelihood fits significantly better. Particularly appropriate for comparing zero-inflated against standard distributions.
- Key Parameters:

- Count Distributions / N Components / Period / Max Iterations / Max Function Evaluations / Clean Data: As in 4.2.
- Selection Test: Criterion for model comparison at each stage. Options: AIC, BIC, Vuong, F. Default: AIC.
- Evaluation Order: Components first (default): selects the best N, then selects the best distribution. Distribution first: selects the best distribution, then the best N.
- Output Interpretation (Results Table):
In addition to all fields from 4.1, the following selection metadata are reported:
 - Selected Count Model: Name of the winning distribution.
 - Selected N Components: Harmonic complexity of the winning model.
 - Selection Test: Criterion used for the automated selection.
 - All Results: The full comparison grid (equivalent to 4.2 output) is retained and available for export, enabling the user to verify the selection.

4.4 Parameter Confidence Intervals

- Purpose: Computing 95% bootstrap confidence intervals for the rhythm parameters (amplitude, MESOR, and acrophase) of a specified count model. Because count-data GLMs do not provide closed-form analytical CIs for derived rhythm parameters — as opposed to raw regression coefficients — this method uses a bootstrap resampling strategy.
- Mechanism: Repeatedly subsamples 2/3 of the observations, re-fits the specified model on each subsample, and collects amplitude, MESOR, and peak estimates across iterations. CIs are constructed as $\text{mean} \pm 1.96 \times \text{SD}$ across bootstrap replicates. Peak matching across iterations uses a tolerance window (Precision Rate) to ensure that peaks from independent fits are attributed to the same acrophase before averaging.
- Key Parameters:
 - Count Distribution / N Components / Period / Max Iterations / Max Function Evaluations / Clean Data: As in 4.1. Should match the model selected in 4.1 or 4.3.
 - Repetitions: Number of bootstrap iterations. Higher values yield more stable CI estimates at the cost of computation time. Default: 20. Recommended for publication: ≥ 50 .
 - Precision Rate: Tolerance in time units (hours) for matching peaks across bootstrap iterations. Peaks within $\pm$ Precision Rate hours of the reference peak are considered the same acrophase before averaging. Default: 2.0 hours.
- Output Interpretation (Results Table):
 - Amplitude CI [lower, upper]: 95% bootstrap confidence interval for the rhythm amplitude.

- MESOR CI [lower, upper]: 95% bootstrap confidence interval for the MESOR.
- Peak CI (h) [lower, upper]: Per-acrophase 95% CI. One row per detected peak. If no peaks are detected across bootstrap iterations (e.g., the fitted rhythm is flat), these fields will be empty.
- Peak Height CI [lower, upper]: Per-acrophase 95% CI for the predicted count value at the peak.
- Count Model / N Components / Repetitions: Metadata identifying the model and bootstrap settings used.

Note on stability: With Repetitions = 20, CIs are approximate. If the lower and upper bounds differ substantially between runs, increase repetitions to ≥ 50 . Computation time scales linearly with this parameter.

4.5 Group Comparison

- Purpose: Characterizing circadian rhythm parameters across multiple experimental groups (conditions, genotypes, treatments) and compiling a comparative summary table. For each group, the best-fitting model is selected automatically and bootstrap confidence intervals are computed, enabling descriptive comparison of amplitude, MESOR, and acrophase across conditions.
- Mechanism: For each unique value of the specified grouping column, the method independently executes the two-stage model selection procedure from 4.3 followed by the bootstrap CI estimation from 4.4. Results are assembled into a single summary table with one row per group. Unlike CircaCompare, which fits a joint model and tests for differential rhythmicity via likelihood ratio testing, this method fits independent models per group — making it appropriate when different groups may require different count distributions (e.g., one group is zero-inflated while another is not).
- Key Parameters:
 - Group Column: Name of the column in the dataset that defines the groups (e.g., condition, genotype, treatment). Must be a categorical variable. Required.
 - Count Distributions / N Components / Period / Max Iterations / Max Function Evaluations / Clean Data: As in 4.2. Applied independently to each group.
 - Selection Test / Evaluation Order: As in 4.3. Applied independently to each group.
 - Repetitions / Precision Rate: As in 4.4. Applied independently to each group.
 - Plot Comparison: If enabled, renders a multi-panel figure with one subplot per group, showing raw data, the best-fit count-cosinor curve, and CI bands. Default: disabled.

- **Output Interpretation (Results Table):**
One row per group. All rhythm parameter fields (Amplitude, MESOR, Peaks, Heights) and goodness-of-fit fields (LLR p-value, AIC, BIC, pseudo-R²) are as defined in 4.1. The following additional fields are included:
 - [Group Column]: The group identifier (value of the grouping variable for that row).
 - Selected Count Model / Selected N Components: Winning model for that specific group. Note that different groups may have different winning models — this is expected and valid when variance structures differ across conditions.
 - Amplitude CI / MESOR CI / Peak CI / Peak Height CI: Bootstrap confidence intervals for each parameter, computed independently per group as described in 4.4.

Interpretation guidance: When comparing groups, focus on whether the confidence intervals for a given parameter overlap between rows. Non-overlapping CIs provide informal evidence of a difference. For formal hypothesis testing of group differences in amplitude or acrophase, use the CircaCompare module (Module III), which provides likelihood ratio-based significance tests.

5. MODULE V: AI CONSENSUS – CONSENSUS RHYTHMICITY SCORE

- **Purpose:** A method-agnostic rhythmicity classifier that integrates evidence from four independent circadian analysis methods into a single probability score (0–1), reducing the dependence on any single algorithm's assumptions.
- **Mechanism:** The module operates as a meta-classifier in three stages:
 - **Feature Extraction:** Five sub-methods are run automatically on the input time series: JTK_CYCLE, Cosinor OLS, Lomb-Scargle Periodogram, and Fourier F24. From their results, 11 features are extracted, including individual p-values, tau statistics, R² values, the F24 score, and cross-method aggregate metrics (method agreement and period concordance).
 - **Machine Learning Prediction:** The feature vector is fed to a pre-trained Random Forest pipeline (SimpleImputer → RandomForestClassifier, 200 trees, max depth 10, balanced class weights). Missing features due to method failure are handled with a sentinel value (–999), which the model has learned to interpret as "feature unavailable" rather than evidence for or against rhythmicity.
 - **Classification:** The model outputs a rhythmicity probability (0–1) and assigns a categorical label based on fixed thresholds: Rhythmic (≥ 0.70), Borderline (0.30–0.70), or Arrhythmic (< 0.30).

- Training data: The model was trained on 3,299 instances with verified ground truth labels: ~1,600 synthetically generated time series spanning a range of periods, noise levels, and waveform shapes, plus ~1,699 real biological profiles from two independent GEO datasets (GSE11923, mouse liver hourly × 48h, Hughes et al. 2009; GSE11516, mouse liver 4h × 48h, with BioCycle labels from RhythmicDB). Classes are balanced during training.
- Key Parameters: None. The meta-classifier runs all sub-methods with internal defaults (period range 18–32h). No user configuration is required or available.
- Output:
 - Probability score (0–1): The primary result. Higher values indicate stronger evidence for rhythmicity.
 - Classification: Rhythmic / Borderline / Arrhythmic based on the fixed thresholds above.
 - Per-method radar chart: Normalized contribution scores for JTK, Cosinor OLS, Lomb-Scargle, F24, Harmonic Cosinor, and overall 4-Method Agreement.
 - Feature importances: Relative contribution of each input feature to the Random Forest's decision, allowing identification of which sub-methods drove the classification.
 - Sub-method summary: Individual p-values and detected periods from each of the five internal analyses.
- Important limitation: The model was trained exclusively on gene expression time series. It is not available for DAM (locomotor activity) data, from which it is automatically hidden.

6. MODULE VI: Locomotor Activity Analysis (DAM/AWD data)

This module is available exclusively when DAM (Drosophila Activity Monitor) or AWD (running wheel, Actimetrics ClockLab) data is loaded. It does not appear for CSV or Rosbash data sources, nor when the data source has not yet been processed.

Data loading prerequisites

- DAM files — ChronoScope uses MultiDAMDataLoader to parse the standard Trikinetics tab-separated format (32 activity channels per file). Configurable parameters:
 - Bin size (min): Temporal resolution of activity counts.
 - Lights ON hour: Defines ZT0 (start of light phase).
 - Exclude dead channels: Removes channels with ≥ N h of consecutive zero activity.
 - Death threshold (h): Consecutive inactivity hours used to classify a channel as dead.

Each DAM file is treated as one condition; the condition name is auto-detected from the filename.

- AWD files — ChronoScope uses MultiAWDDataLoader to parse the Actimetrics ASCII format (single-column wheel revolution counts with header). Configurable parameters: bin size, lights-on hour (ZT0), date range, and missing-bin handling. Each AWD file is treated as one subject; condition names are set by the user.

6.1. Activity Profile

- What it does: Performs a comprehensive characterization of locomotor activity across all loaded conditions. A single run generates six visualization tabs plus a summary metrics table, covering daily patterns, temporal rhythmicity, period estimation, and behavioral metrics.
- Key Parameters:
 - LD ends day: Last recording day of the light-dark (LD) phase. Set to 0 if the recording starts directly under constant conditions. Default: 0 (none).
 - DD ends day: Last day of the constant-darkness (DD) phase. Set to 0 to extend DD to the end of the recording. Default: 0 (end). The adjacent skip first N days field excludes the initial N days of DD from the τ calculation; the first DD day is excluded by default (default: 1) as it is typically a transition day from LD.
 - τ search (h) / α : Period search range for the Sokolove–Bushell chi-square periodogram, and the significance level (α) used to classify each individual as rhythmic. Only individuals whose peak Qp exceeds the χ^2 threshold at the chosen α are included in the τ estimate. Default: 18 – 32 h, $\alpha = 0.05$.
 - α/p threshold: Threshold method for classifying active vs. rest bins: mean uses the daily mean activity; percentile-25 uses the 25th percentile of that day's activity distribution. Default: mean.
 - Normalize each subject to its total activity: When enabled, each subject's activity at every time bin is divided by its total activity across the entire recording before any averaging is performed. This ensures that all subjects contribute equally to the group-averaged profile regardless of their absolute activity level. This normalization applies to the Activity Profile heatmap, Actogram, Daily Average, Onset/Offset, Chi-square Periodogram, IS/IV, and α/p computations. It does not affect the Total Daily Activity chart. Default: off.
 - Onset (min bins / k): Controls onset and offset detection. Min bins sets the minimum number of consecutive bins that must exceed the threshold for a bout to count as an onset or offset event — single-bin spikes are ignored (default: 3 bins). k defines the threshold as mean +

$k \cdot \sigma$ of that individual's activity on that day; lower values yield an earlier onset/wider active window (default: $k = 0.5$).

Phase logic: ChronoScope infers phase boundaries from the three "ends day" values. If LD ends at day 7 and DD ends at day 0, the LD phase covers days 1–7 and DD covers days 8 through the end of the recording. The chi-square periodogram is computed exclusively on the DD phase; if no DD phase is defined, all available data are used.

Bin-size handling: The bin size is inferred automatically from the loaded data. For DAM recordings with bins smaller than 30 minutes, the time series is resampled to 30-minute bins before computing the chi-square periodogram to keep computation tractable. IS and IV are always computed on 1-hour bins regardless of the native resolution.

- Outputs — six visualization tabs:

Tab 1: Heatmap

Displays a day $\times$ ZT-time activity matrix for each condition as a color-coded heatmap. Rows represent recording days; columns represent ZT (Zeitgeber Time) from 0–24 h in bins matching the loaded data resolution. The colormap (YlOrRd) is normalized globally across all conditions for direct cross-group comparison. A light/dark phase indicator bar is shown above each heatmap.

Tab 2: Daily Average

Shows the mean $\pm$ SEM activity profile averaged across all days and subjects for each condition. The X-axis spans ZT 0–24 h; the light phase (ZT 0–12 h) and dark phase (ZT 12–24 h) are indicated by background shading. One line per condition is plotted with shaded error bands.

Tab 3: Actogram

Displays a classic double-plotted actogram (0–48 h per row) for each condition. Activity is represented as vertical black bars with height proportional to normalized activity counts. Each row corresponds to one recording day, with the following day's data repeated in the second 24 h block. The X-axis is labeled in ZT hours (ZT0–ZT48), where ZT0 always corresponds to lights-on regardless of clock time. LD shading is applied independently to the left half (current day) and right half (next day) of each row, so phase transitions appear on the correct side.

If lighting phases are specified in the parameters, a colored sidebar is drawn on the right edge of the actogram marking the LD (yellow), DD (dark), and LL (pale yellow) recording phases by day. Phase labels appear next to each band.

Tab 4: Total Activity

Plots total daily activity (sum across all subjects) as a function of day number, with mean $\pm$ SEM error bands. One line per condition. This panel summarizes activity level trends across the recording period.

Tab 5: Onset / Offset

Displays daily activity onset and offset times (in ZT hours) for each condition. Onset and offset are computed per individual subject for each recording day. For each subject, a day-specific threshold of $\text{mean} + k \cdot \sigma$ is calculated (where k is user-configurable); the onset is the start of the first run of $\geq N$ consecutive bins above threshold, and the offset is the end of the last such run (N and k are set in the Onset parameters). Individual onset and offset times are then averaged across subjects; the plot shows mean $\pm$ SEM across individuals. Onset is shown as filled circles with error bars; offset as open triangles with error bars. Lines connect consecutive days. Days where no individual produces a valid onset or offset are excluded.

Tab 6: Chi-sq Periodogram

Computes the Sokolove–Bushell (1978) chi-square periodogram to estimate the free-running period τ . The periodogram is run per individual subject on the DD phase (skipping the configured number of initial transition days). Each individual is classified as rhythmic if its peak Q_p exceeds the χ^2 significance threshold at the chosen α level. Only rhythmic individuals contribute to the final estimate.

The plot shows the mean Q_p curve $\pm$ SEM shading across rhythmic individuals as a function of period (h). A dashed vertical line marks τ , annotated as mean $\pm$ SEM of individual τ values (e.g., $\tau = 23.84 \pm 0.21$ h) along with a rhythmicity count (e.g., $n = 12/16$ rhythmic). A red dotted horizontal line marks the Q_p significance threshold; its numerical value is shown at the right edge of the plot. A note in the title indicates whether the periodogram was computed on the DD phase or on the full recording.

Results table: Displays one row per condition summarizing all computed metrics:

- n : Number of subjects.
- Days: Total recording days analyzed.
- τ (h): Mean $\pm$ SEM of individual free-running periods, estimated from the chi-square periodogram across rhythmic subjects only.
- τ sig. ($p < 0.05$): Rhythmicity count for that condition (n rhythmic / n total). Cells where at least one subject is rhythmic are highlighted green; all-arrhythmic conditions are highlighted red.
- Phase (τ): Recording phase on which the periodogram was computed (DD or all)
- IS (Interdaily Stability): Ratio of the variance of the mean 24 h profile to total variance (1 = perfectly consistent across days; 0 = random). Computed on 1 h bins.
- IV (Intradaily Variability): Ratio of the variance of consecutive-bin differences to total variance. High values indicate fragmented, bursty activity. Computed on 1 h bins.
- α (h): Mean duration of the active phase across days, defined as the proportion of bins per day exceeding the selected threshold $\times 24$ h.
- α SD (h): Day-to-day variability of α .

- ρ (h): Mean rest phase duration ($\rho = 24 - \alpha$).
- ρ SD (h): Day-to-day variability of ρ .
- Onset SD (h): Day-to-day standard deviation of the mean daily onset time — a measure of onset variability across the recording.
- Offset SD (h): Day-to-day standard deviation of the mean daily offset time.